ScienceDirect

Journal of Manufacturing Processes

Date: 26 December 2025

Show more

Research article Full text access Get rights and content

Mechanisms of femtosecond laser plasmonic micro-welding between sapphire and rough SiC ceramics with natural stacking using ultrafast optical imaging

Changhao Ji [a b], Ziyue Yu [a b], Jindou Wu [a b], Qijian Zhu [a b], Chengyun Wang [c d], Liming Lei [e], Hongchao Qiao [f], Yu Long [a b] Show more

View PDF Download full issue Cite Add to Mendeley Share 10.1016/j.jmapro.2025.10.112

Article Recommended articles Cited by (1) Metrics

Outline

Abstract

Femtosecond laser micro-welding (FSLMW) of ceramics has emerged as a highly promising technology for applications in aerospace, medical devices, and optical sensing. However, achieving reliable bonding remains challenging due to stringent optical contact requirements and unclear underlying mechanisms, especially under practical conditions. To overcome these limitations, this paper introduces a femtosecond laser multi-scan micro-welding approach to directly join sapphire and rough silicon carbide (SiC) ceramics under natural stacking conditions, without the need for optical contact or external forces. Pre-scanning enhances laser absorption, facilitating thermal accumulation that expands the laser-ceramic interaction zone and increases the molten material volume. This molten material effectively bridges macro-scale gaps, enabling successful femtosecond laser microwelding of naturally stacked ceramics. This method can greatly reduce the requirements for ceramic clearance and surface roughness, and overcome the limitations of optical contact. The plasma evolution mechanism at the ceramic interface was investigated using an ultrafast pump-probe shadow imaging system, which provided real-time insights into the dynamic interaction between laser-induced plasmas and material behavior. Furthermore, this paper systematically explores the impact of key welding parameters (laser power, scanning velocity, and the number of laser passes) on the microstructure and mechanical properties of the joints. Notably, the presence of ceramic nanoparticles within the heterojunction contributes to enhanced bonding strength. Additionally, the formation of laser-induced periodic surface structures (LIPSS) on the SiC surface was found to further improve the bonding strength, achieving shear strengths as high as 30MPa. By correlating plasma evolution with microscopic morphology and fracture behavior, this research provides a comprehensive understanding of the plasma dynamics and welding mechanisms, offering valuable insights into the uncertainty surrounding the underlying mechanisms of ultrafast laser micro-welding for ceramics with contrasting physical properties.

Keywords

Femtosecond laser micro-welding; Natural stacking; Sapphire; Rough SiC ceramic; Plasma evolution; Ultrafast optical imaging

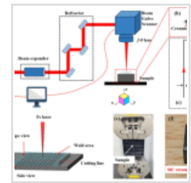
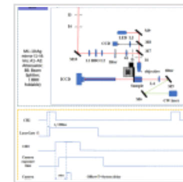
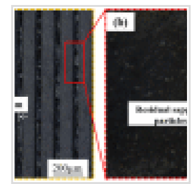
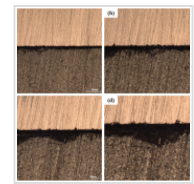
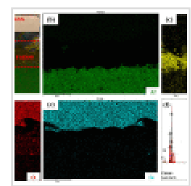
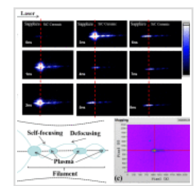
Previous article in this issue

Next article in this issue

1. Introduction

Sapphire, the single-crystal form of Al_2O_3, is renowned for its exceptional mechanical strength, as well as its outstanding resistance to wear and corrosion, making it a vital dielectric material across various industries [[1], [2], [3]]. It is extensively used in laser

Figures (17)

Show 11 more figures

Extras (1)

Supplementary figures

technology, electronics, medical devices, and aerospace applications, particularly as an optical window material [4,5]. Its transparency and durability also make it suitable for use in transparent armor systems [6]. Silicon carbide (SiC) ceramics, offering superior ballistic performance compared to alumina and achieving approximately 70–80% of the protection capability of boron carbide, are particularly attractive for large-scale use due to their relatively low cost. These characteristics make SiC ideal for lightweight, thin protective armor applications [7,8]. As a next-generation trend in transparent bulletproof materials, ceramic-based armor remains challenged by several key issues. A primary limitation lies in the insufficient size of individual transparent ceramic plates, necessitating reliable joining techniques for splicing. However, the inherent brittleness and processing difficulties of ceramics significantly hinder their broader adoption and raise safety concerns. Traditional joining methods, such as brazing and diffusion bonding, typically require interlayers to promote interfacial reactions [[9], [10], [11], [12]]. These reactions often generate new phases at the interface, increasing residual stress. Moreover, such methods demand complex processing conditions, including high temperatures, high pressures, and controlled atmospheres, leading to increased time and cost. Consequently, these techniques are inadequate for the emerging requirements of high-performance, low-damage, and localized joining in miniaturized sapphire/ceramic assemblies, particularly in applications like bulletproof armor materials.

Recent advancements in ultrafast laser micro-welding, particularly femtosecond laser technology, have shown great promise in joining dissimilar brittle materials with minimal heat-affected zones. This technique has been successfully applied to various material combinations, such as glass-glass [13], glass-metal [[14], [15], [16], [17], [18], [19]], semiconductor-metal [20], glass-semiconductor [[21], [22], [23], [24]], and ceramic-ceramic [25], achieving shear strengths comparable to traditional methods while avoiding issues like aging and degassing. As a result, femtosecond laser micro-welding (FSLMW) has emerged as a promising alternative for high-precision, high-durability bonding applications. Pioneering work by Itoh et al. [[26]] first demonstrated the use of femtosecond lasers for ceramic sealing. Subsequently, Garay et al. [25] further advanced the field by employing ultrashort pulse lasers (USPL) to weld yttria-stabilized zirconia (YSZ) and polycrystalline alumina. By optimizing heat treatment conditions, they enhanced the near-infrared transparency and nonlinear absorption of YSZ, thereby improving both its functional properties and bond strength. This exemplifies a key advantage of FSLMW: the ability to simultaneously join and tailor materials for specific applications, such as electronic packaging. Building on this, Pan et al. [19] successfully bonded sapphire and Fe—36Ni alloy using FSLMW, achieving defect-free joints with interface widths under 1 μm. Key to this success were surface polishing and the use of compression fixtures to establish optical contact. Yuan et al. [27] also optimized process parameters to bond alumina ceramics, achieving joints with bending strengths up to 60% of the base material's strength. This result is significant because it shows that FSLMW can produce strong joints without compromising material performance. In another study, Jiang et al. [28] joined $MgAl_2O_4$ and Ti6Al4V by fine-tuning laser power and scanning speed, thereby optimizing both microstructure and shear strength. Collectively, these studies demonstrate that ultrashort pulse laser welding holds great promise for ceramic bonding. Its ability to achieve high-strength joints with minimal heat input, coupled with its adaptability to different materials and applications, makes it a highly efficient and versatile bonding method. As the technology continues to evolve, further optimization of laser parameters will likely expand its applicability, particularly in demanding fields such as aerospace, electronics, and armor materials.

In summary, after over a decade of development, femtosecond laser micro-welding (FSLMW) has achieved significant progress, with shear strengths in some cases approaching that of the base material. Despite these advances, several challenges still hinder its industrial application. A major issue is the requirement that the contact gap between the samples must not exceed one-quarter of the wavelength [29], or even be within 100 nm [30], in order to achieve stable and reliable welds. Achieving such conditions demands not only mirror-like surface polishing but also the application of external pressure to maintain intimate contact during welding. This requirement poses significant challenges for practical applications. The need for fixtures complicates the welding process and can introduce internal stress when the samples relax post-welding, leading to mechanical tension at the seam, crack formation, and reduced joint integrity. While several studies have attempted to address the limitations of optical contact in transparent material welding, the resulting weld strength and quality often remain insufficient for industrial-scale deployment. Therefore, overcoming the challenges of welding naturally stacked samples is critical to advancing the engineering applicability of FSLMW. Currently, non-optical contact FSLMW techniques typically rely on single-scan strategies, overlooking the potential benefits of multi-pass scanning. Initial scans can significantly improve the laser absorption coefficient, potentially exceeding 90%, while also facilitating heat accumulation for subsequent scans. Moreover, multi-pass strategies can mitigate early-stage defects such as ablation-induced damage and thermally induced cracks, resulting in improved weld quality and structural integrity. Although efforts have been made to optimize structural design, processing workflows, and laser energy modulation to enhance joint strength, further investigation is required to better understand the interfacial micro-welding mechanisms under non-optical contact conditions.

In this paper, we demonstrate the successful direct micro-welding of sapphire and rough SiC ceramics with natural stacking using multi-scan FSLMW technology, eliminating the need for surface polishing or pressure fixture assistance. This method greatly reduces the requirements for glass contact gap and surface roughness and flatness. Additionally, plasma splashing during the sapphire-ceramic welding process was captured with an ultrafast pump-probe shadow imaging system, providing valuable insights into the welding mechanism. By precisely optimizing laser energy deposition at the interface, we achieved sapphire/SiC ceramic joints with unique microstructures and high shear strength. We systematically examined the effects of laser power, scanning speed, and the number of welding passes on the microstructure and shear strength of the joints. The mechanisms behind the formation of high-quality joints under varying laser parameters were analyzed, leading to the identification of optimal welding conditions. This paper introduces femtosecond laser welding as an innovative and efficient approach for sapphire-ceramic joining, offering superior precision compared to conventional welding techniques.

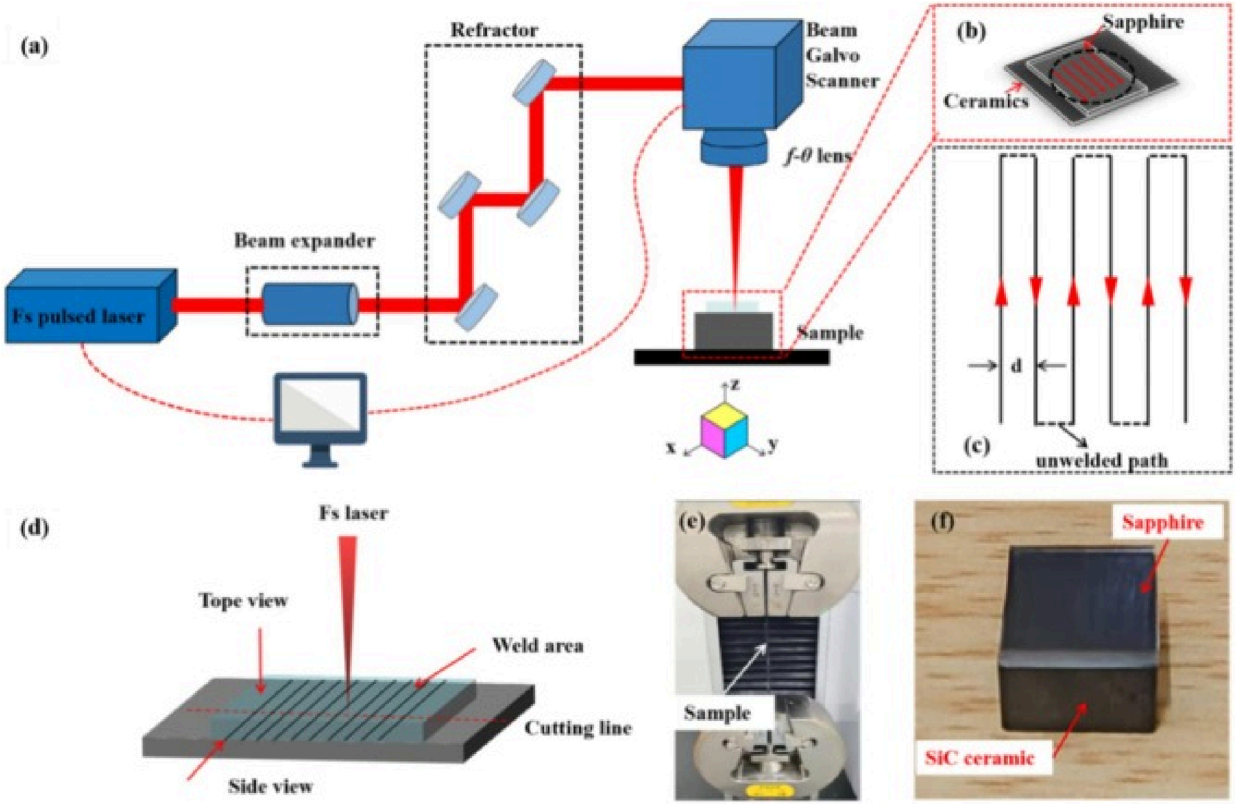
2. Experiment procedure

2.1. Materials

The sapphire utilized in the experiment was supplied by the professor Neri's laboratory., while the ceramic used was the rough SiC ceramic fabricated by the Guangzhou Baile New Material Technology Co. Both the SiC ceramic and sapphire were precisely cut into $10\times10\times4\,mm^3$ and $10\times10\times1\,mm^3$, respectively. Specifically, the sapphire substrates were purchased with a flatness of λ/4, ensuring high optical quality, while the SiC ceramics exhibited a surface roughness of $Ra=1–2\,\mu m$ prior to welding. The samples were cleaned with a solution of acetone and alcohol under ultrasonic vibration, followed by drying. Before the femtosecond laser welding experiment, the glass and ceramic were wiped clean with alcohol to remove any dust, fingerprints, or other dirt from the surfaces. The sapphire and SiC ceramic were naturally stacked together under a non-optical contact condition. “Non-optical contact (NOC)” refers to a condition in which a micrometer-scale gap exists between the two materials to be welded during the process. The material surfaces exhibit micrometer-scale roughness (as in the case of the SiC ceramic in this study) and are therefore not perfectly flat, smooth, or tightly bonded. In contrast, “Optical contact” requires the surface gap to be smaller than one-quarter of the laser wavelength, which typically demands ultra-precision polishing and the use of alignment fixtures. Non-optical contact represents a more practical and commonly encountered configuration in industrial applications. Under this condition, the surfaces are sufficiently close to allow laser-induced plasma expansion and localized melting to bridge the interface, yet no van der Waals or optical interference contact is established.

2.2. Femtosecond laser welding system

The schematic diagram of the femtosecond (fs) laser welding experimental setup is shown in Fig. 1(a). The fs laser used in this experiment is the FemtoYL-40 femtosecond laser provided by Wuhan Anyang Laser Technology Co., Ltd. This laser has an output wavelength of 1030 nm (invisible light), adjustable pulse width of 300–6000 fs, and a maximum power of 40 W. The repetition rate is adjustable from 0.025 to 1 MHz. The femtosecond laser focus was concentrated at the interface between the glass and metal. After setting the appropriate laser parameters, the y-axis was accelerated, and when the set speed was reached, the laser was emitted. The sapphire and SiC ceramic welding was achieved by scanning in a linear pattern. To ensure optimal welding quality, the laser was turned off before completing the uniform scan, preventing any potential effects from platform acceleration and deceleration. Fig. 1(b) illustrates the laser scanning path during the welding process. The scan pattern consisted of parallel lines with a perpendicular spacing of 100 μm. To assess the mechanical properties of the joints, the shear strength was tested using an electronic universal testing machine, as shown in Fig. 1(c). The loading rate was set to 0.5 mm/s, and at least three specimens were tested to ensure the reliability and consistency of the experimental results. Both lateral and fractured surface observations of the welding area were conducted. The preparation process for lateral observations is shown in Fig. 1(d). The welding area was placed at the edge of the sample, embedded, ground, and then observed laterally (Side view). The samples were polished using 2000-grit sandpaper and a final polish with a 0.05 μm alumina suspension. For fractured surface observations, the samples were separated, and the surface morphology of the welding area was observed (Top view), as shown in Fig. 1(d).



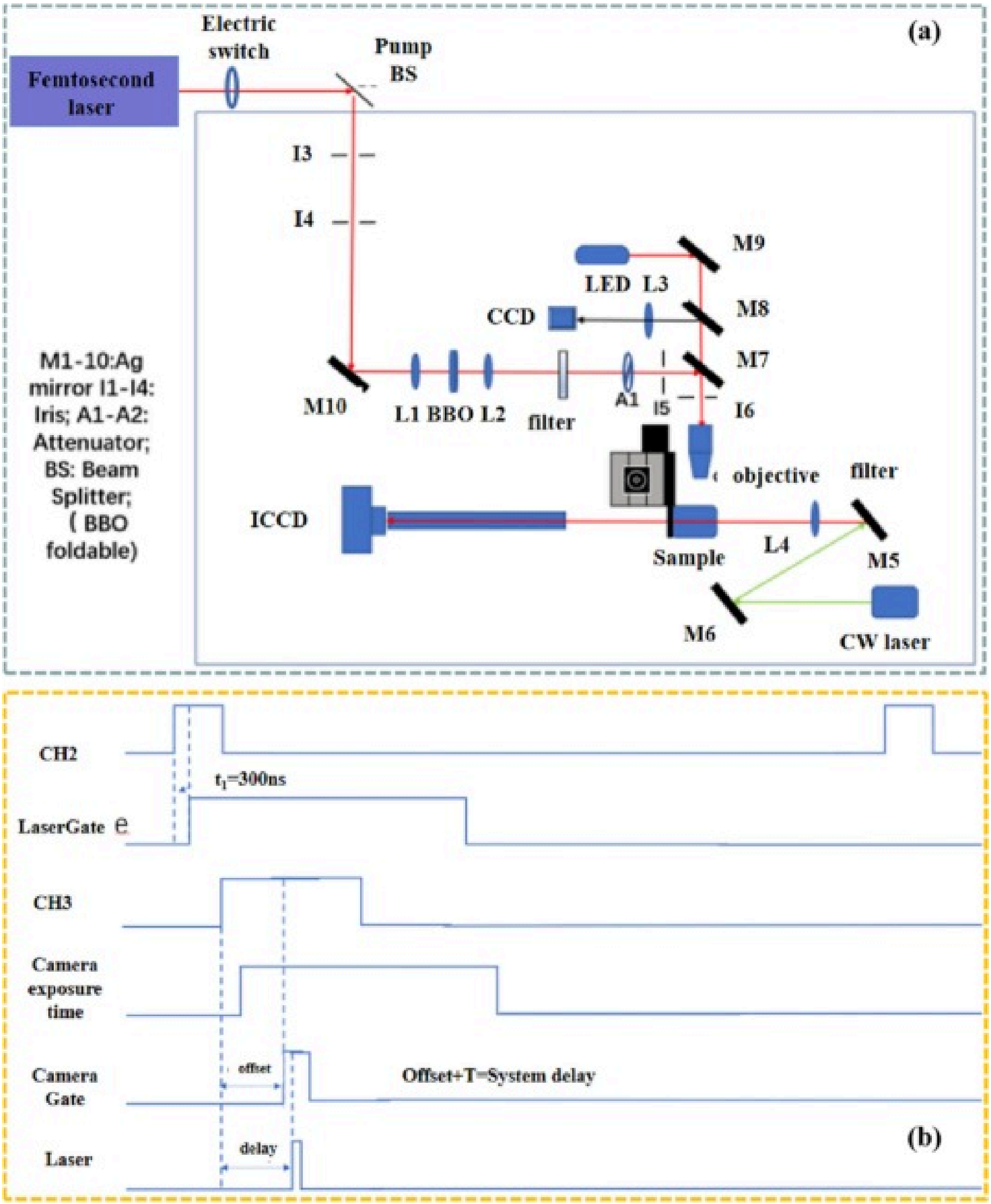
Download: Download high-res image (156KB)

Download: Download full-size image

Fig. 1. The schematic diagram of the femtosecond laser micro-welding(FSLMW) system (a) Schematic diagram of the FSLMW system and Scanning path; (b) samples; (c) scanning trajectory; (d) The assemble of sapphire and SiC ceramic samples; (e) shear strength test; (f) Typical macroscopic morphology of fs laser-welded sapphire/SiC ceramic.

2.3. Ultrafast pump-probe shadow imaging system

To monitor the plasma during the glass-ceramic welding process, we constructed an ultrafast pump-probe shadow imaging system. The ultrafast pump-probe detection shadow imaging device (TACCD-100) is depicted in Fig. 2(a). In the experiment, we utilized an amplified titanium-sapphire laser system, producing pulses of 35 fs duration at a repetition rate of 1 KHz, with a central wavelength of 800 nm. The TACCD-100 mainly consists of a laser source, mechanical delay unit, BNC-2121 junction box, three-dimensional electric displacement stage, beam splitter, and imaging detector (ICCD camera). The plasma splashing test path is illustrated in Fig. 2(b), wherein the camera's gating function and delay are utilized to capture the plasma splashing process at different times. The entire testing process involves focusing the pump on the sample, which generates plasma. At this point, the camera's exposure time is fixed, acting as a “gate,” and the delay relative to the pump hitting the sample is adjusted through timing to obtain images of the plasma splashing process at different times (the testing process can be referenced in Fig. 2(b) timing diagram). This function can also perform long-term (ns) testing. Additionally, a 532 nm cw laser can be used as an illumination source for testing, allowing detection of changes in plasma splashing at different delay times. This technique employs a femtosecond laser pulse as the pump light to excite the sample, generating plasma on the surface. The ICCD camera receives the reflected signal from the sample surface, and by adjusting the time delay between the camera and pump light, images of the evolving plasma at different times are recorded. Analyzing changes in the images provides insight into the evolution process of the plasma, including propagation speed and attenuation patterns.



Download: Download high-res image (322KB)

Download: Download full-size image

Fig. 2. The schematic diagram of the ultrafast pump-probe shadow imaging system (TACCD-100) (a) Optical path of the ultrafast pump-probe shadow imaging system for plasma testing; (b) Timing diagram for shadow imaging.

2.4. Analysis of microstructure and mechanical properties of joint

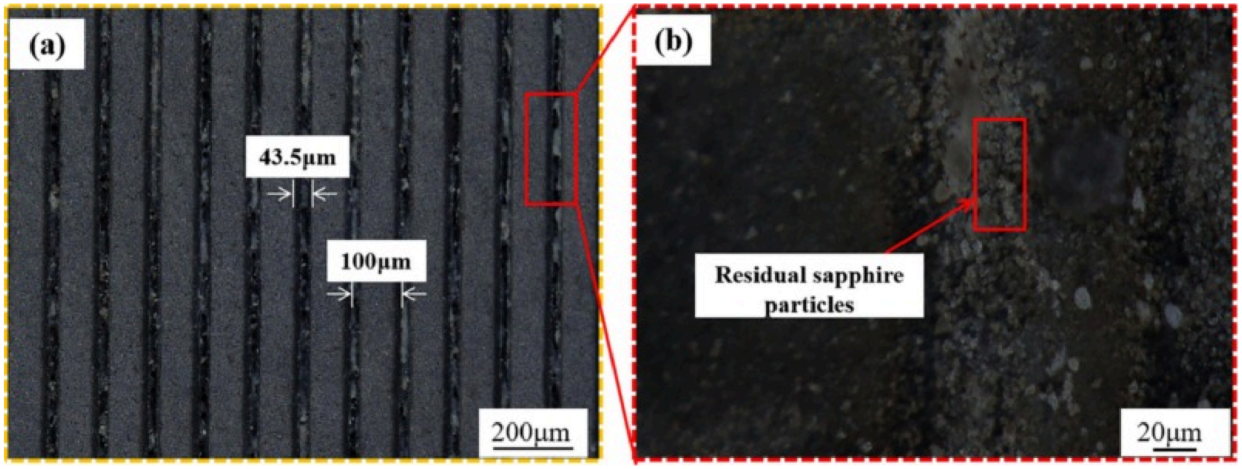
All experimental samples were subjected to tensile-shear tests using a universal testing machine. Based on the principle of close numerical values, the obtained tensile stress values were grouped, and one sample was selected from each group for simple metallographic analysis, marking important positions for scanning electron microscope (SEM) scanning. The welded specimens were placed on the electronic universal testing machine for joint shear strength testing. In the welding performance test, the welded area was longitudinally cut and polished, and the microstructure of the sapphire-SiC ceramic interface was observed using a scanning electron microscope (SEM), with the interface element distribution analyzed using energy-dispersive X-ray spectroscopy (EDS).

3. Results and discussion

In this study, sapphire and SiC ceramics were cleaned with acetone and anhydrous ethanol to ensure surface purity, without any polishing treatment. This approach mitigates the typical challenges of FSLMW, such as the strict requirements for minimal gaps, material uniformity (optical contact), and alignment over large areas. By varying process parameters (laser power, scanning speed, number of scanning), the optimal welding conditions for sapphire-ceramic connections were identified. In this method, a femtosecond laser pulse with low energy density is focused on the contact surface between sapphire and SiC, then the laser focusing spot moves along the small area rapidly and multi-scan repeatedly drived by a two-dimensional scanning galvanometer. The absorption coefficient of the laser can be increased by the previous scan, and a certain amount of heat will be accumulated before the next scan, which can expand the area of laser-ceramic interaction, which is conducive to the increase of the melting volume of the target material to obtain enough molten material to fill the large gap, thus successfully realizing the femtosecond laser microwelding of naturally stacked ceramics. This method can greatly reduce the requirements for ceramic gap and surface roughness, and reduce the limitation of optical contact. Although no optical contact exists between sapphire and rough SiC surfaces, reliable bonding is still achieved. This is because plasma-induced melting and nanoparticle ejection at asperity contacts generate molten bridges that span across micro-gaps. The confined plasma plume exerts transient high pressures, which drive molten material to infiltrate interfacial voids and establish physical contact. In addition, multi-scan irradiation promotes cumulative thermal accumulation, enabling progressive expansion of the bonded region. Therefore, the femtosecond laser multi-scan process achieves bonding under practical stacking conditions by relying on plasma-mediated bridging and gap filling, rather than atomically smooth interfaces.

3.1. Microstructures and elemental diffusion behavior analysis of weld joint

Femtosecond laser pulses with a frequency of 0.2 MHz, a duration of 300 fs, a wavelength of 1030 nm, and a pulse count of 1 were used to directly weld sapphire glass and SiC ceramics. The microstructure of the weld seam is shown in Fig. 3(a), where no cracks were observed in the irradiated region. The welding line width was approximately 43.5 μm, with the centers of adjacent welding lines spaced about 100 μm apart. A magnified image of the SiC ceramic residuals is provided in the upper-right corner of Fig. 3(a). After shear-force separation, Fig. 3(b) reveals a significant amount of dark-colored sapphire residue on the SiC ceramic surface, indicating the formation of a strong bond between the sapphire and SiC ceramics in the non-optical contact area. This suggests that a robust joint has been established between the sapphire and SiC ceramics within the irradiated region.

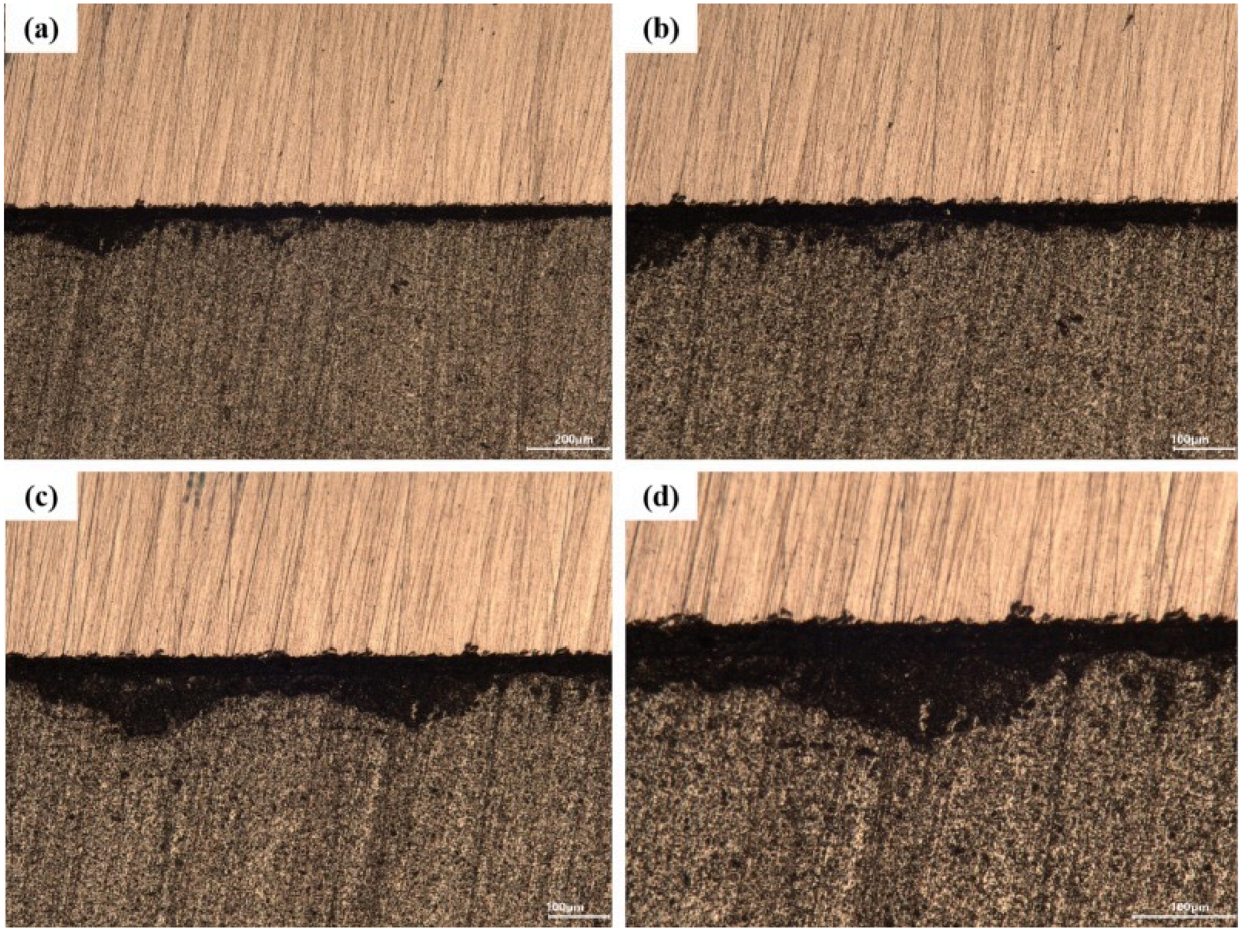


Download: Download high-res image (453KB)

Download: Download full-size image

Fig. 3. (a) Sapphire-SiC ceramic welding line; (b) Enlarged view of the red box in (a). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The internal gap between the materials is significantly reduced after femtosecond (fs) laser irradiation, as shown in Fig. 4(a). The high peak power of the fs laser induces a series of nonlinear absorptions in the sapphire when focused on the interface between sapphire and ceramic. The resulting rapid heat buildup in the confined space hinders heat dissipation, leading to the melting and even gasification of the materials at the focal point. Once the molten sapphire and ceramic combine, a black bonding layer, as seen in Fig. 4(a), is formed. The average width of this bonding layer is approximately 50 μm, which likely enhances the mechanical strength of the joint. However, as depicted in Fig. 4(b–d), without external clamping pressure, some plasma escapes during the welding process. If the plasma diffuses too far from the molten pool, it can reduce the bonding effectiveness of the molten zone.

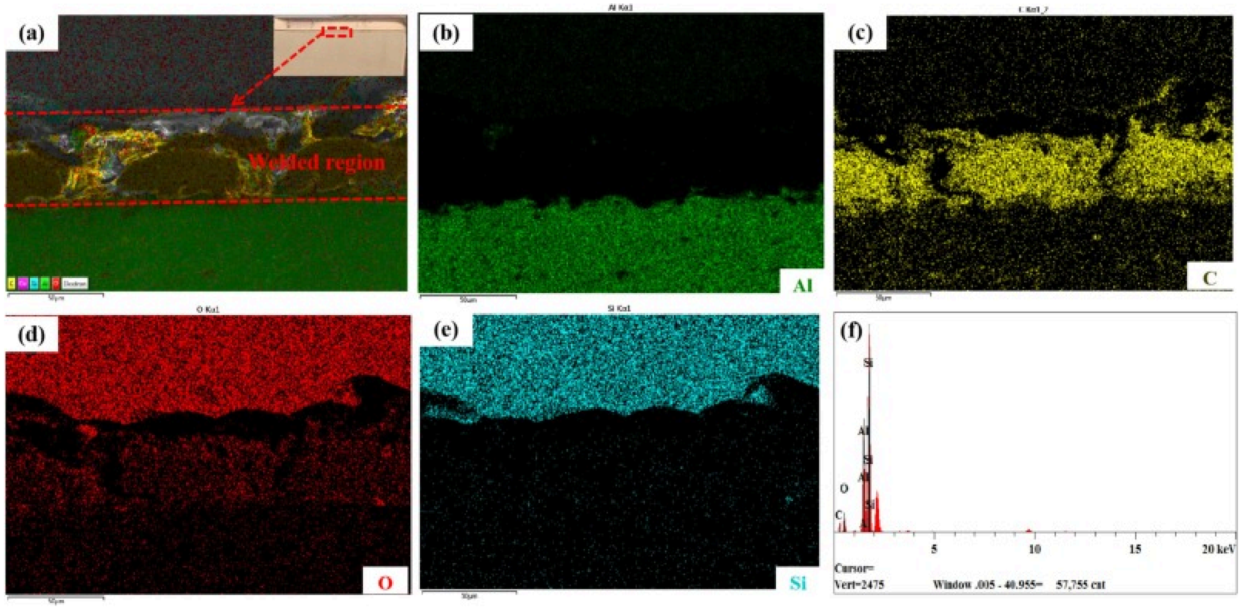


Download: Download high-res image (2MB)

Download: Download full-size image

Fig. 4. Microscopic appearance of Cross-section of weld area: (a) Black bonding layer (side view); (b–d) Conversion zone.

Next, surface analysis using SEM and EDS was performed to obtain SEM images and element distribution within the weld seam cross-section, as shown in Fig. 5. Fig. 5 presents a full view of the cross-section of the sapphire/SiC ceramic joint welded with femtosecond laser under non-optical contact conditions. As shown in Fig. 5(a), periodic welded zones with irregular shapes are visible on the sapphire side near the sapphire/SiC ceramic interface. The spacing between adjacent welded zones is approximately 100 μm, corresponding to the spacing between the scanning paths during the welding process. From both Figs. 4(a) and 5(a), it is evident that each welded zone parallel to the interface is about 45 μm wide, which is significantly larger than the 20 μm diameter of the femtosecond laser focus. The height of the welded zones perpendicular to the interface generally does not exceed 100 μm, indicating that the welded zones are located near the interface and do not extend deeply into the sapphire. Fig. 5(b–e) show the element distribution of O, Si, and Al on the surface of the glass-ceramic joint. The main elements present in the welding area are Al, O, Si, and C. Elemental diffusion at the sapphire glass and SiC ceramic interface in the weld seam suggests material mixing and mutual diffusion during the laser irradiation. The strong oxygen signal detected in the upper SiC region (Fig. 5d) can be attributed to oxidation and material mixing during femtosecond laser welding. Under intense plasma and localized heating, partial decomposition of Al_2O_3 releases oxygen, while SiC at high temperature readily oxidizes to SiO_2. These oxygen species migrate into the SiC side and form SiO_2 or mixed Al-Si-O compounds at the interface. Compared with the uniform lattice oxygen in bulk sapphire, these newly formed oxides exhibit higher local oxygen concentration and stronger EDS contrast, which explains the dominant O signal observed in the upper region. As shown in Fig. 5(e), a few island-like Si-rich areas are dispersed within the welded zones, indicating the incorporation of Si into the welded regions.



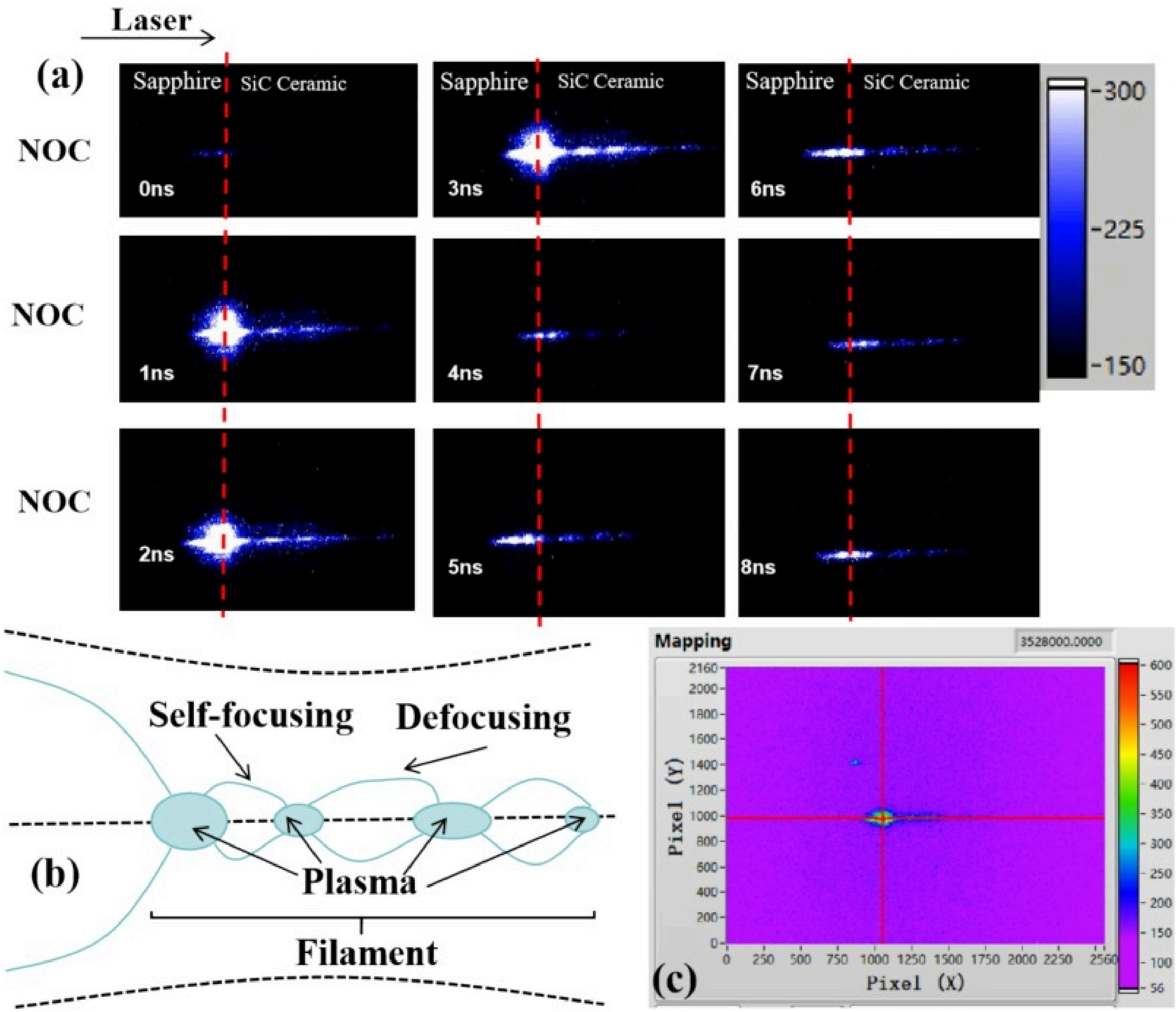
Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 5. (a) The typical microstructure of Sapphire/SiC ceramic joint welded by femtosecond laser; (b–e)The EDS elemental distribution at the interface of sapphire-SiC ceramic joints; (f)Energy spectrum of the laser-sapphire interaction region.

3.2. Plasma images of the interaction area with natural stacking

The interaction of femtosecond laser at the without optical contact (WOC) interface of sapphire/ceramic was studied using the pump-probe technique, which is commonly used in recording and analyzing the transient evolution for both the time and space domains on materials surfaces. This paper primarily utilizes the gate acquisition function of an ICCD camera with delay to capture different moments of the sample's plasma splashing process. The testing procedure involves focusing the pump on the glass-ceramic sample's contact surface, generating plasma, with fixed exposure time serving as a “gate.” Adjusting the sequence relative to the laser's impact time enables capturing various plasma splashing processes. The evolution process of the plasma for glass at different delay times is depicted in Fig. S1 (Supporting Information). We investigated the process of single laser pulse welding focused on the sapphire-ceramic interface, directly observing the plasma excitation process on the nanosecond time scale (Fig. 6(a)). Filamentation of femtosecond laser pulses is observed during ultrafast laser propagation in a medium. In glass mediums, when the laser's peak power surpasses the self-focusing threshold, Kerr effect-induced self-focusing balances with plasma generation's self-defocusing effect, leading to filamentation and characteristic plasma channel generation involving multiphoton ionization and avalanche ionization. However, excessive incident power may result in multiple filaments due to pulse modulation instability, leading to a stochastic process of filament generation with random distribution of the optical field's spatial mode. The images of plasma and shock waves reveal the longitudinal enhancement effect of plasma channels on shock wave expansion and the influence of expansion dimensions. The duration of the entire process is short, and phenomena such as plasma and shock waves gradually weaken and disappear after 3 ns. Within the delay time of 1 ns to 3 ns, we observed the presence of weak plasma not only on the sapphire-ceramic interface but also on the ceramic interface. This indicates that the self-focusing phenomenon of glass triggers secondary initiation on the ceramic interface, resulting in the generation of weak plasma, albeit with minimal energy.



Download: Download high-res image (554KB)

Download: Download full-size image

Fig. 6. (a)Evolution process of plasma at different delay times and the same position on the sapphire-ceramic interface; (b) Schematic diagram of the self-focusing phenomenon; (c)The time-resolved images of plasma generated by single-pulse irradiation at different positions on the glass-ceramic interface (2 ns).

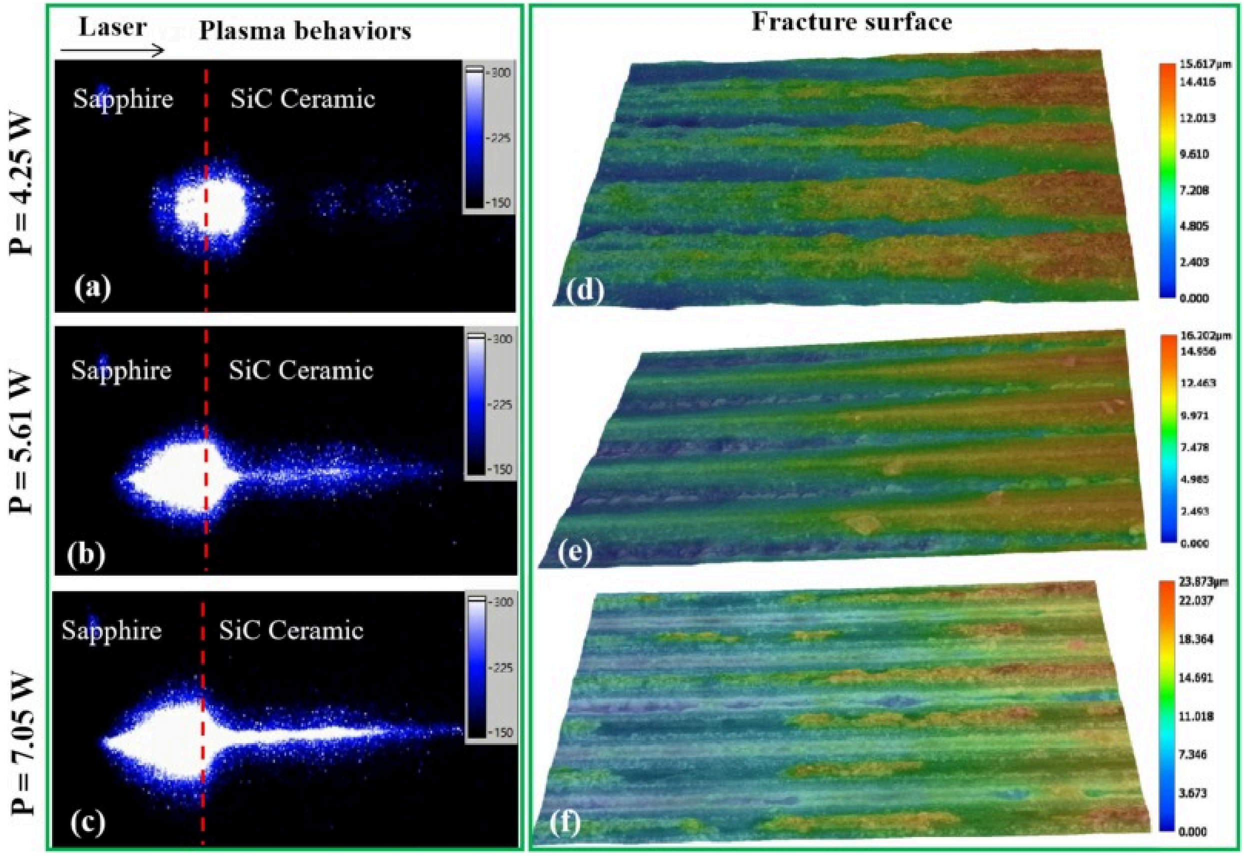
Furthermore, the plasma on the ceramic interface disappears only after a delay time of 8 ns. According to Braun et al. [31], when the peak power of the ultrafast laser exceeds the self-focusing threshold, the beam can overcome the diffraction effect and undergo self-focusing propagation. As shown in Fig. 6(b), after self-focusing of the ultrafast laser, ionization is excited in a self-guiding manner, generating plasma with a certain lateral density gradient. As depicted in Fig. 6(c), the dynamic balance between self-focusing in the solid and defocusing effect of plasma leads to an uneven spatial distribution of filamentous plasma [32]. The self-focusing and nonlinear absorption in sapphire glass are key factors affecting the shear strength because energy absorption and modification in sapphire glass can induce internal stress and reduce energy deposition at the contact interface.

Next, we investigated the time-resolved images of plasma generated by single-pulse irradiation of the glass-ceramic interface with femtosecond excitation (0–7 ns) and the corresponding evolution of the plasma (Figs. S2 and S3, Supporting Information). The duration of the entire process was very short, with the phenomenon of plasma and shockwave gradually diminishing after 3 ns until only slender filamentary plasma remained at the end. Within the delay time of 1–3 ns, it was observed that besides the plasma at the glass-metal interface, there was also a weaker presence of plasma at the ceramic interface. This suggests that the self-focusing phenomenon of the glass led to a secondary ignition at the ceramic interface, resulting in the generation of relatively weak plasma despite its low energy. Plasma was observed at 1 ns, with the signal intensity peaking at 2 ns. After 3 ns, the plasma gradually diffused until it completely disappeared at 7 ns, with the entire process lasting approximately 8 ns. Although the plasma lifetime was measured to be on the order of ~3 ns, the interval between successive pulses at a repetition rate of 0.2 MHz is ~5 μs, much longer than the plasma dissipation time. This means that direct pulse–plasma interactions are negligible under our experimental conditions. Instead, the laser–material interaction proceeds through a cumulative effect: each pulse generates transient plasma, followed by localized lattice heating, defect generation, and permanent refractive index modification. These residual changes, persisting into the microsecond regime, act as “seeds” that alter the optical and thermal environment for the following pulse. Consequently, while plasma itself does not overlap temporally with subsequent pulses, the cumulative effects from multiple pulses significantly influence the energy coupling efficiency, morphology evolution, and bonding quality at the sapphire/SiC interface. The difference between multi-pulse action (Fig. S4, Supporting Information) and single-pulse action (Fig. 6(a)) is that under single-pulse action, the dynamic balance between the self-focusing effect of the glass and the defocusing effect of the already generated plasma leads to an uneven spatial distribution of filamentary plasma. In contrast, under multi-pulse action, the filamentary plasma is distributed relatively evenly in space, and with the increase in the number of pulses, both the uniformity and the length of the plasma filaments gradually increase. This is mainly due to the reduction of the defocusing effect of the already generated plasma under multi-pulse action.

In 1995, scientists at the University of Michigan, led by Braun, achieved plasma channels with a transmission distance of 20 μm and nearly constant energy in air using femtosecond lasers. This plasma channel, known as filamentation, is the result of the interaction between high-power ultrafast lasers and Kerr-transparent media. The filamentation process involves rich nonlinear optical effects and can ionize the transmitting medium. During propagation through transparent sapphire, femtosecond laser pulses can undergo nonlinear self-focusing due to the Kerr effect once the peak power exceeds the critical threshold. This phenomenon effectively reduces the focal spot size at the sapphire/SiC interface and increases the local laser intensity. As a result, plasma formation is enhanced and energy deposition becomes more confined, which facilitates localized melting and bonding. The self-focusing effect also helps to partially overcome interfacial roughness and micro-scale gaps by ensuring that sufficient energy is concentrated directly at the joint region.

As shown in Fig. 7(a–c), under the same number of pulses, the length of the plasma inside the sapphire glass first increases and then levels off. This indicates that the separation of plasma inside the sapphire glass is limited due to strong nonlinear absorption.

Additionally, the intensity and height of the plasma increase with the laser power, indicating that sapphire glass absorbs more energy, leading to increased internal modification and stress. Fig. 7(d–f) illustrates the fracture surface morphologies (top view) of sapphire joints at different laser power. For the joint welded at 4.25 W, the 3D topographies revealed that the fracture surfaces on both the sapphire and SiC sides exhibited numerous bumps and holes. Specifically, pit-like structures were observed on the sapphire side (Fig. 7(d)), primarily resulting from the melting and solidification of sapphire near the interface. Additionally, the fracture surface contained multiple holes with dispersed SiC nanoparticles. These features suggest that fracture preferentially occurred along the weak interfacial regions with inherent defects. At a higher laser power of 5.61 W, the fracture surface displayed distinct pits and protrusions (Fig. 7(e)). The step-like structures observed are characteristic of brittle fracture, consistent with the hard and brittle nature of the sapphire substrate. When the laser power was increased to 7.05 W, the joint exhibited a clean fracture along the weld seam on the sapphire side, with large SiC ceramics fragments (23.837 μm in height) remaining adhered to the SiC ceramics (Fig. 7(f)).

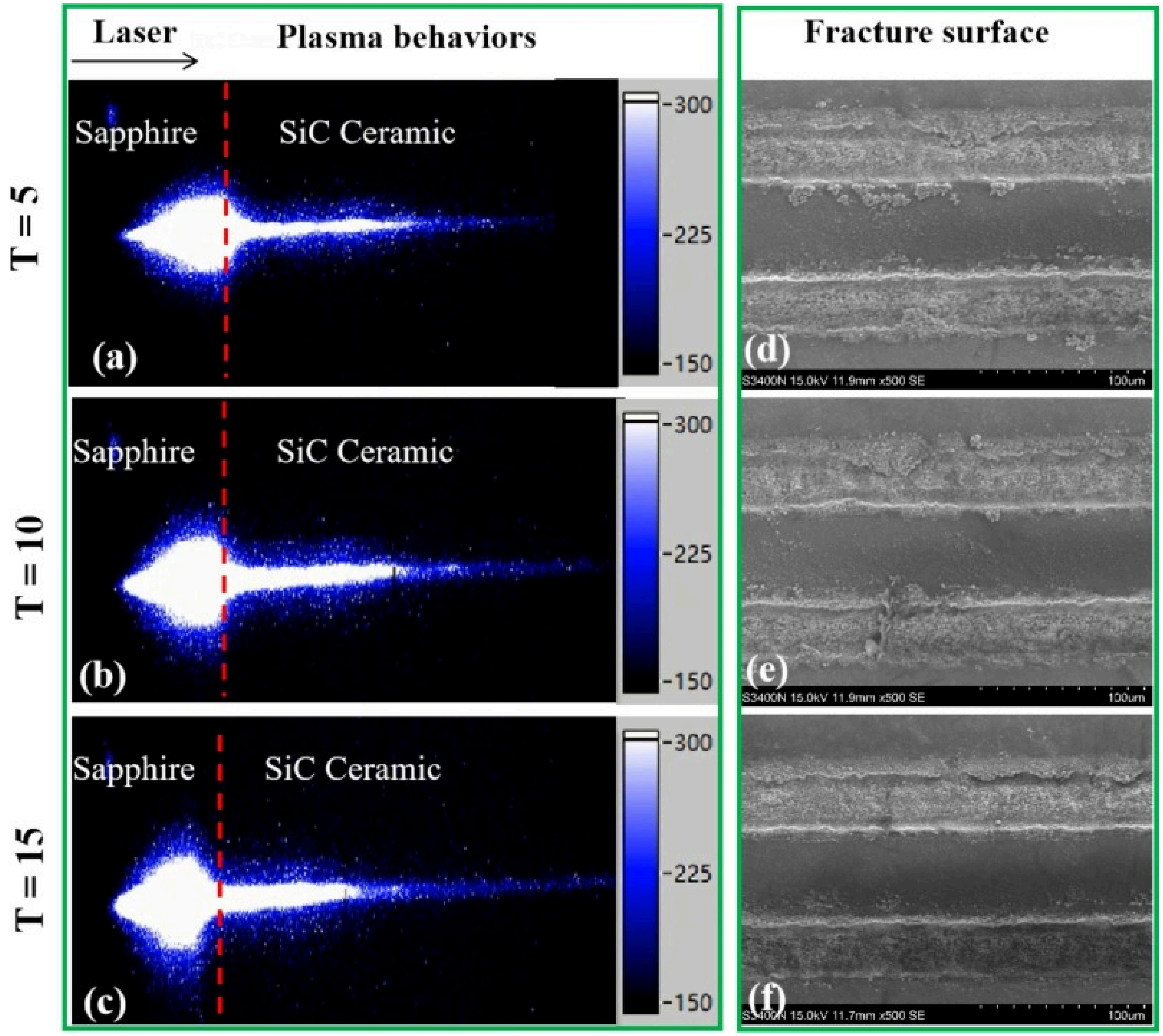


Download: Download high-res image (771KB)

Download: Download full-size image

Fig. 7. Plasma behaviors at different laser power of 4.25, 5.61, 7.05 W (a–c) Evolution process of plasma on the sapphire-ceramic interface at different laser power (P=4.25, 5.61, 7.05 W); (d–f) 3D topography of sapphire fracture surface of the joints welded at different laser power (P=4.25, 5.61, 7.05 W).

Fig. 8 presents plasma behaviors and SEM images of the weld fractures obtained under different number of welding. As shown in Fig. 8(a–c), under the different number of welding, with the increase in the number of welding, filamentation phenomena are observed at the glass-ceramic interface, and the length of the filaments increases with the number of pulses, along with an increase in the brightness of the plasma filaments. Laser transmission exhibits significant divergence due to diffraction effects, resulting in a short transmission distance and a sharp decrease in energy density. Fig. 8(d–f) illustrates the fracture surface morphologies (top view) of sapphire joints at different number of welding. Following the separation of sapphire from ceramic, residual micro/nano-sized particles were observed on the sapphire-ceramic interface. These particles, which serve as adhesive remnants during the welding process, indicate high welding quality.



Download: Download high-res image (809KB)

Download: Download full-size image

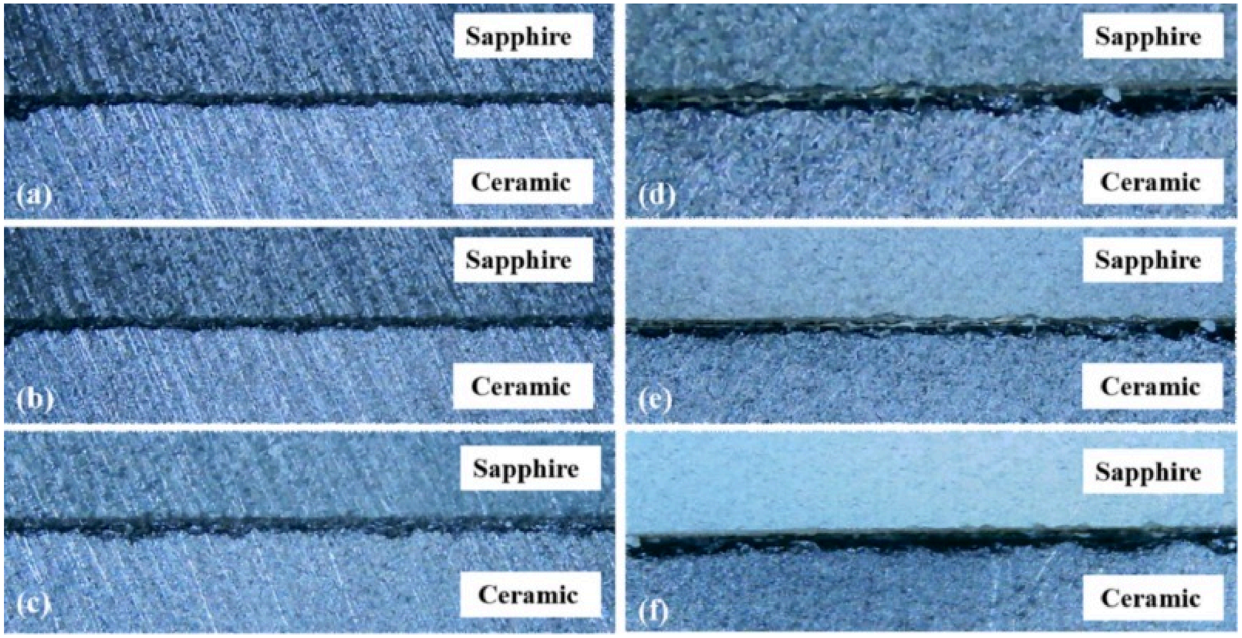
Fig. 8. Plasma behaviors and SEM images of weld fracture at different number of welding (a–c) Evolution process of plasma on the sapphire-ceramic interface at different number of welding (T=5, 10, 15); (d–f) SEM images of sapphire fracture surface of the joints welded at different laser power (T=5, 10, 15).

Based on the above analysis, the bonding mechanism between the sapphire and SiC ceramic interface under femtosecond laser irradiation can be categorized into four modes: 1) Melting of SiC Ceramic Only: The thermal effect causes the SiC ceramic to melt while the sapphire remains solid. The connection between the two materials is achieved through the cooling and solidification of the molten SiC ceramic; 2) Simultaneous melting of both materials: Both materials reach their respective melting points under laser irradiation. The molten materials are stirred and mixed within the molten pool, facilitating bonding between the two materials; 3) Plasma Formation on SiC Ceramic: Under sufficient laser energy density, the surface of the SiC ceramic generates plasma, which then mixes with the molten sapphire, promoting bonding; 4) Plasma Interaction at the Interface: Both materials at the interface produce plasma, which mixes and diffuses. Due to the selective absorption of 1030 nm wavelength laser by both sapphire glass and SiC ceramic, the laser beam directly interacts with the oxide layer on the SiC ceramic surface through the sapphire, leading to material mixing and diffusion. The mixed material near the interface serves as a buffer, effectively mitigating the mismatch in thermal expansion and melting points between the glass and ceramic. This buffering effect is a key factor in achieving reliable bonding between glass and ceramic.

The observed changes in surface morphology and particle formation can be attributed to the interplay between plasma generation and laser penetration depth. Under femtosecond laser irradiation, the intense plasma generated at the interface produces ultrahigh temperature and pressure, resulting in transient microexplosions. These processes drive material ejection and splashing, which are responsible for the formation of nanoparticles and micro-debris near the welded region. A higher degree of plasma excitation generally leads to rougher surface features and a larger amount of molten particle release. At the same time, the penetration depth of the laser determines the spatial extent of plasma propagation: deeper penetration promotes subsurface plasma expansion, which may induce localized cracks or voids, whereas shallow penetration confines the plasma to the near-surface region, yielding finer debris and relatively smoother morphologies. Therefore, the interdependence between plasma dynamics and penetration depth plays a decisive role in controlling the microstructural and morphological characteristics of the welded interface. Although the pump–probe test was performed with 35 fs, 800 nm pulses, the observed plasma initiation and decay dynamics are representative of the 300 fs, 1030 nm welding condition, since both operate in the femtosecond regime. The results confirm that plasma shielding is transient (~3 ns) and that each pulse interacts independently with the interface, providing qualitative insight into the bonding mechanism.

3.3. Optimization of sapphire-SiC ceramics welding process

Due to the femtosecond laser's pulse width being significantly shorter than both the electron-phonon coupling time and the material's thermal diffusion time, heat diffusion is minimal throughout the laser's duration, resulting in a small heat-affected zone. The laser's extremely high peak power and low average power allow for precise energy input control, minimizing the risk of cracking during sapphire processing. Laser power, welding speed, and the number of welding directly affect the quality of the weld seam and joint strength. Fig. 9(a–c) show the metallographic images of the weld seam interface at different laser powers, where the thickness of the black bonding layer increases with the increase of power parameters. Upon femtosecond laser irradiation of the ceramic surface, nanoscale particles are initially generated at the pulse site, which then rapidly melt and merge with the molten sapphire, forming a black bonding layer between the sapphire and SiC ceramics. As the laser energy passes through the glass, it is absorbed by the metal layer at the interface, generating heat that melts both the upper and lower materials, which then rapidly cool to form the weld seam. The melting of sapphire and SiC results in an interlocking structure or the formation of a new transition layer. This indicates that when the laser power parameters reach a certain level, a shock wave is formed at the center of the focal point, reducing the material density in the center of the weld seam and increasing it in the surrounding area, resulting in the formation of the molten material.



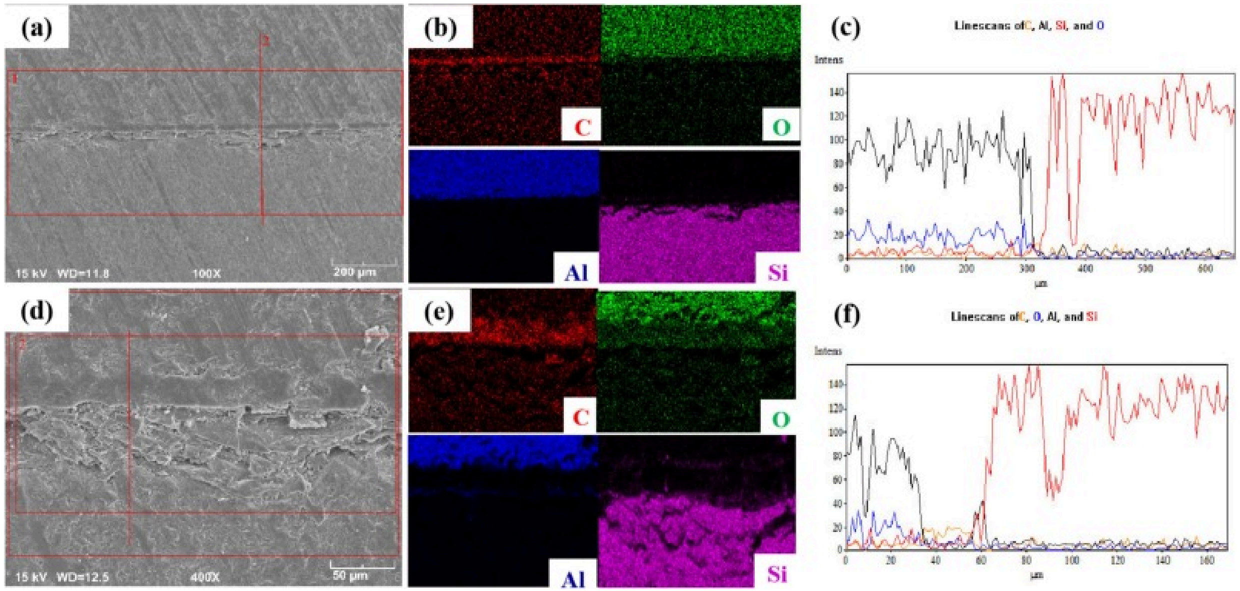
Download: Download high-res image (873KB)

Download: Download full-size image

Fig. 9. Typical macroscopic morphologies of femtosecond laser-welded sapphire/SiC ceramics joints at different laser fluences: (a) P=3.15W; (b) P=4.25W; (c)P=7.05W; (d)V=10mm/s; (e) V=50mm/s; (f)V=100mm/s.

The microstructures of the cross-sections of sapphire/ceramic interfaces joined by femtosecond laser at different scanning speeds are shown in Fig. 9(d–f). In Fig. 9(d), the sapphire/ceramic joints exhibit a morphologically interlocking structure when the scanning speed is 10mm/s. As the scanning speed increases to 50mm/s and 100mm/s, Fig. 9(e–f) reveal that the microstructure of the joints becomes smoother. This change is attributed to the fact that, at higher scanning speeds, each pulse interacts with relatively unmodified regions of the sapphire/SiC interface, producing short-lived plasma plumes that contain contributions from both materials. Although fewer pulses overlap per spot and the overall heat input is lower, these compositionally mixed plasma plumes alter the local energy coupling and lead to smoother bonding zones. The term “number of welding” refers to the number of times the femtosecond laser processes the same welding area, influencing the amount of heat absorbed by the material. When only one pass is made, insufficient heat input leads to a weak and loosely bonded joint. However, as the number of passes increases, the heat input at the same position increases, resulting in a tighter and stronger weld.

Microphotographs of the weld seams are shown in Fig. 10. Laser power is a vital parameter in the process of USPL welding. Fig. 10(a) and (d) display the microstructures of the joints obtained under different laser power with the scanning speed of 10 mm/s. When the laser power was set as 5.61 W, the joint presents a good morphology. It can be found that the interface bonding area, that is, the welding zone, has the effect of pinning to the SiC ceramics side, especially with the laser power of 7.05 W. Next, surface analysis was conducted using SEM and EDS to obtain SEM images and element distribution within the weld seam cross-section at P=5.61 W. Fig. 10(a) presents the characteristic microstructure of femtosecond laser-welded sapphire/SiC ceramic joints, demonstrating effective bonding between the dissimilar materials. The interfacial morphology exhibits distinctive undulating features, resulting from the interaction between the focused femtosecond laser beam and the material interface. This surface modification suggests localized melting and resolidification processes within the laser focal volume. Elemental mapping analysis (Fig. 10(b–c)) reveals the presence of Al, O, Si, and C as the primary constituents in the weld zone. Notably, the distribution patterns indicate the formation of discrete, island-like Si-rich domains within the welded region, providing evidence of silicon migration across the interface during the welding process. There exists a transition layer between the distribution of Al elements and O, Si elements in the glass. Additionally, Al elements are distributed below the upper boundary of the upper layer, while O and Si elements are distributed above the lower boundary of the lower layer. Elemental diffusion occurs at the interface of sapphire glass and SiC ceramic in the weld seam, indicating material mixing and mutual diffusion during laser irradiation.



Download: Download high-res image (878KB)

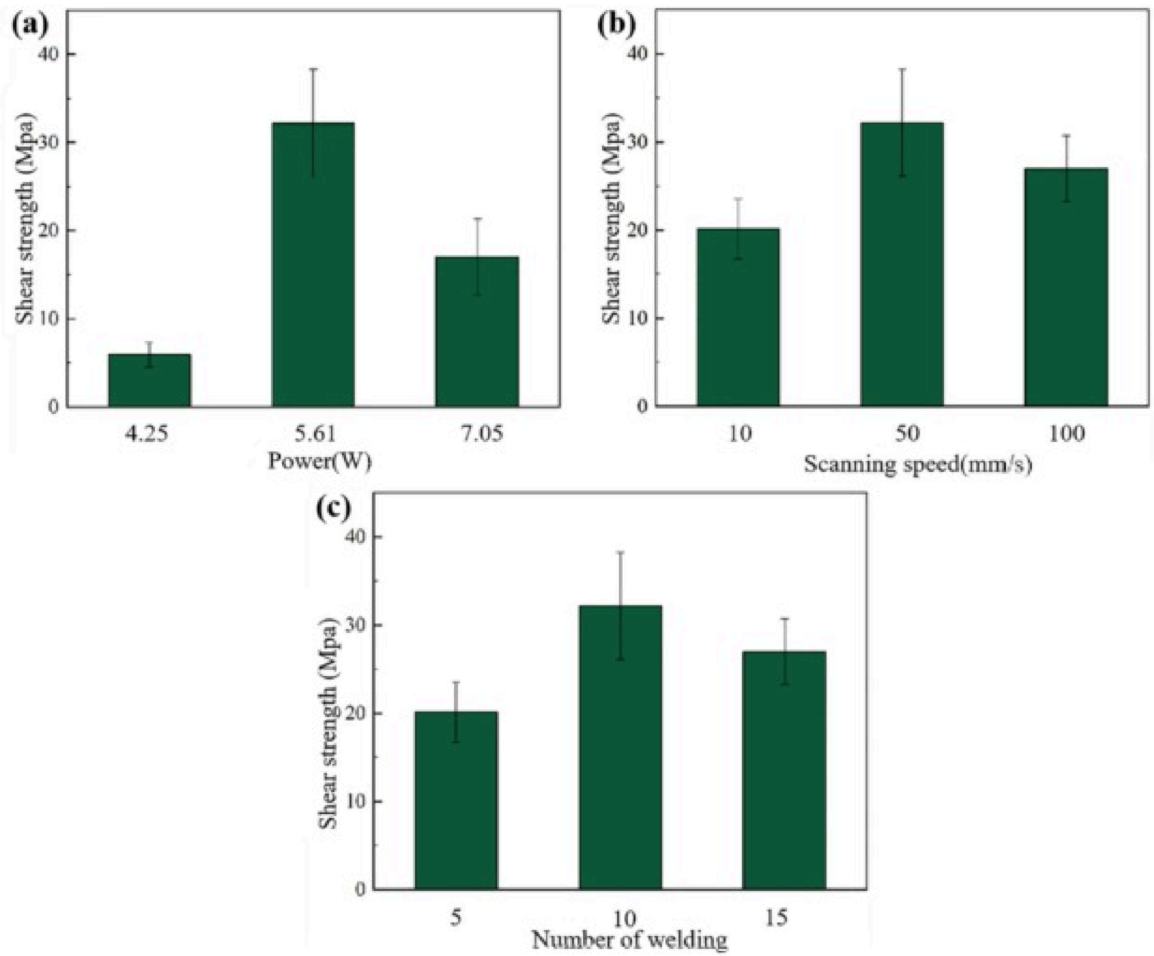
Download: Download full-size image

Fig. 10. (a) The typical microstructure of Sapphire/SiC ceramic joint welded by femtosecond laser at P=5.61 W; (b)The EDS elemental distribution at the interface of sapphire-SiC ceramic joints; (c) EDS line scanning results; (d) The typical microstructure of Sapphire/SiC ceramic joint welded by femtosecond laser at *P*=7.05 W; (e)The EDS elemental distribution at the interface of sapphire-SiC ceramic joints; (f) EDS line scanning results.

Microstructural characterization was performed using SEM coupled with EDS to analyze the weld seam cross-section at a laser power of 7.05 W. Fig. 10(d) presents the representative microstructure of femtosecond laser-welded sapphire/SiC ceramic joints. At elevated laser power levels, microcrack formation was observed on the ceramic substrate, accompanied by a reduction in the pinning effect within the weld zone. Elemental mapping analysis (Fig. 10(e)) reveals the spatial distribution of oxygen, silicon, and aluminum across the glass-ceramic interface. The presence of a continuous transition layer between aluminum-rich and silicon/oxygen-rich regions indicates significant interdiffusion and material mixing during the laser welding process. Furthermore, the identification of discrete Si-rich domains within the weld zone (Fig. 10(f)) provides evidence of silicon migration and redistribution, suggesting the formation of a complex multiphase structure at the interface.

3.4. Mechanical properties analysis of weld joint

To investigate the welding quality of the sapphire/SiC ceramic joint under different welding parameters, the mechanical properties are characterized by shear strength. Fig. 11(a) presents the shear strength of the joint at various laser power levels. It is evident that the shear strength remains below 20 MPa at power levels of 4.25 W and 7.05 W. Specifically, the shear strength is only 5.9 MPa at 4.25 W and increases to 17 MPa at 7.05 W. This suggests that low energy results in ineffective welding, while high energy leads to the formation of interface cracks, both of which hinder the improvement of the joint's mechanical properties. At a power of 5.61 W, the joint reaches its maximum shear strength of 32 MPa. A similar trend is observed with varying scanning speeds and number of welding, as shown in Fig. 11(b–c). At a scanning speed of 10 mm/s, the joint exhibits the lowest shear strength, followed by 100 mm/s. However, since the effect of scanning speed is less significant than that of power, the shear strength remains relatively stable with changes in scanning speed. This behavior can be attributed to competing mechanisms of interfacial bonding enhancement and thermal stress accumulation, where insufficient energy input leads to inadequate bonding while excessive energy causes microstructural degradation.



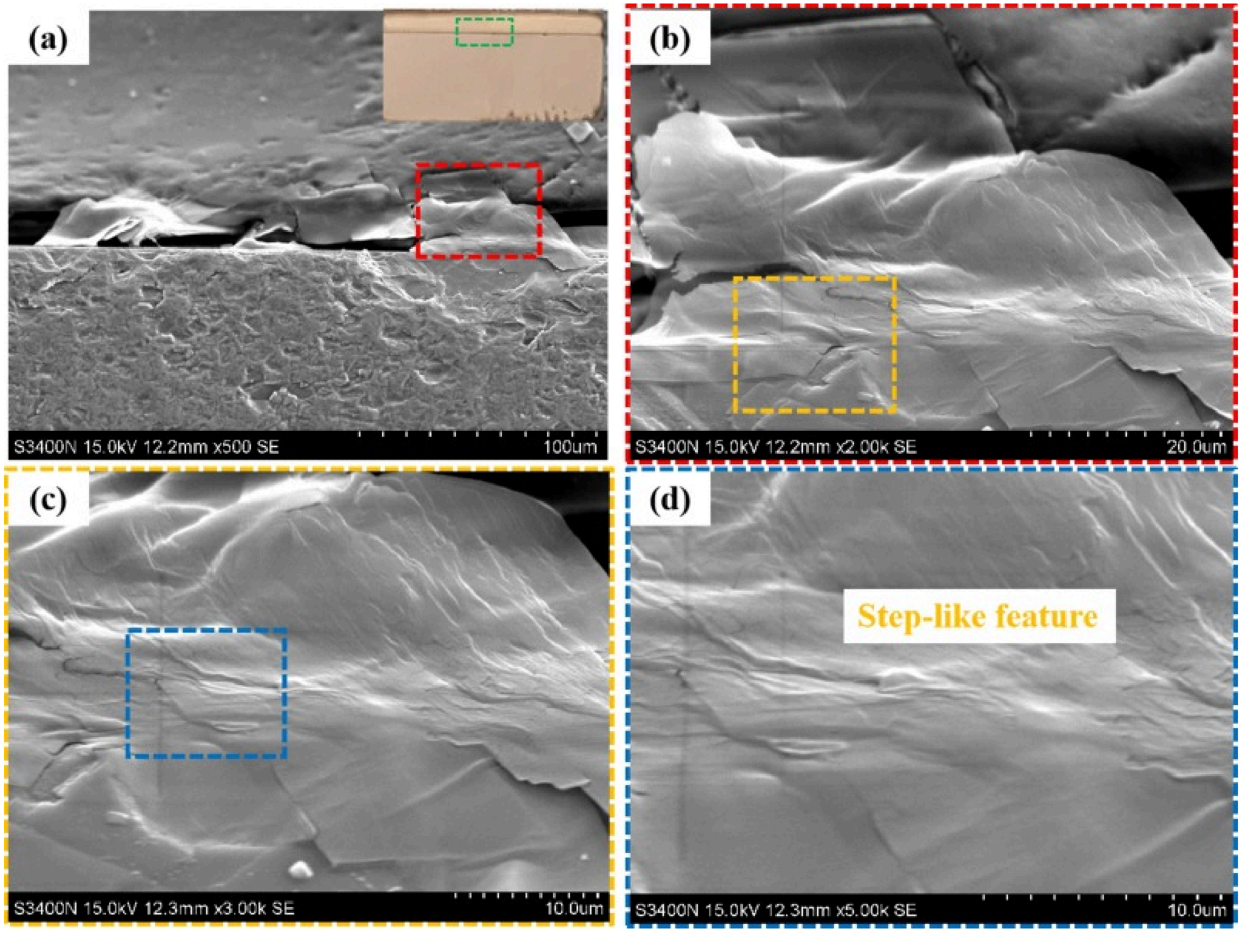
Download: Download high-res image (159KB)

Download: Download full-size image

Fig. 11. Shear strength of Sapphire/SiC ceramics joint with different parameters: (a) laser power; (b) scanning speed; (c) number of welding.

3.5. Characterization of fracture interface

As shown in Fig. 1(d)**,** both lateral and fractured surface observations of the welding area were conducted. Fig. 12 shows the typical fracture interface morphologies of sapphire/SiC ceramic joints after shear strength tests (side view). It was observed that steplike structures with the characteristics of brittle fracture (Fig. 12(b–d)). Figs. S5 and S6 (Supporting Information) presents the SEM images of the weld fractures obtained under different laser parameters. Following the separation of sapphire from ceramic, residual micro/nano-sized particles were observed on the sapphire-ceramic interface. These particles, which serve as adhesive remnants during the welding process, indicate high welding quality. As depicted in Fig. S5 (Supporting Information), the fracture primarily occurs in two distinct regions: the interface reaction zone between sapphire and the oxide layer, and the junction between the oxide layer and the ceramic matrix. Fig. S6(b–d) (Supporting Information) reveal the presence of abundant nanoparticles located between the sapphire stripes on the fracture surface. Compositional analysis of these particles, as depicted in Fig. S6(d–j) (Supporting Information), indicates that they consist of a mixture of SiC and Al_2O_3 nanoparticles.

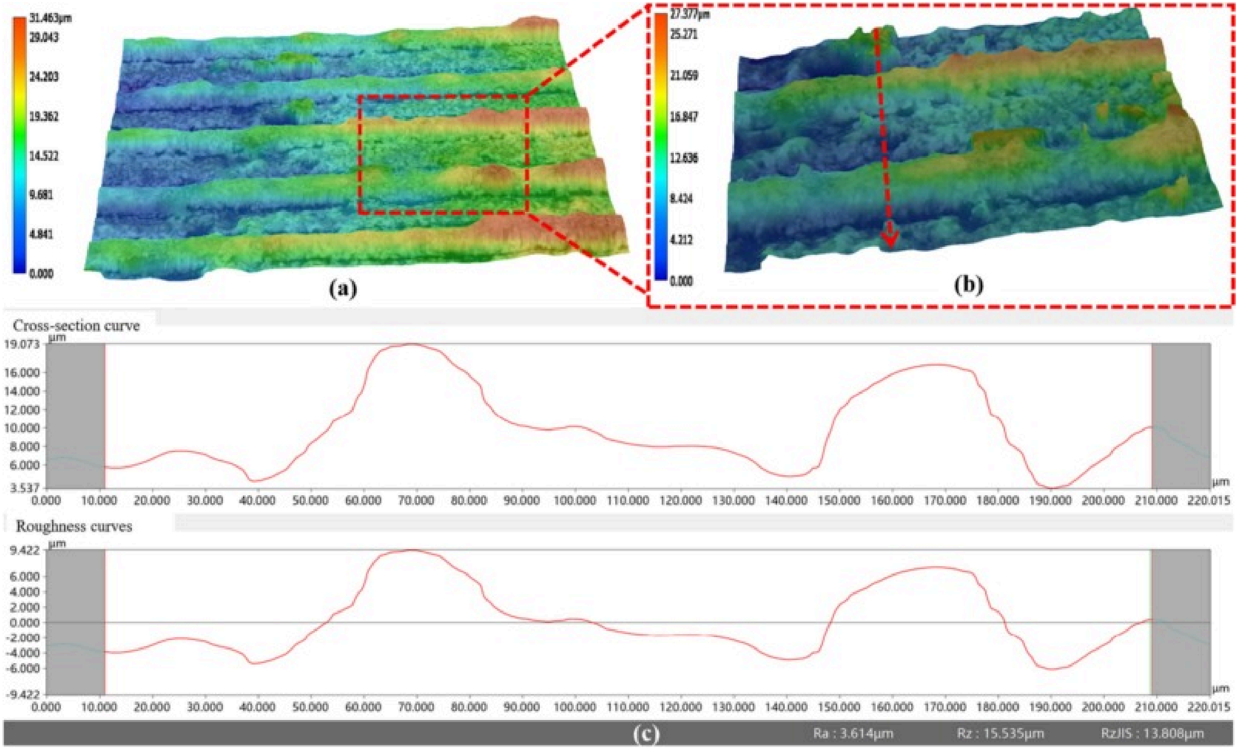


Download: Download high-res image (817KB)

Download: Download full-size image

Fig. 12. The typical fracture interface morphologies of sapphire/SiC ceramic joints after shear strength tests (side view) (a) Fracture interface morphologies of Sapphire-SiC ceramic welding line; (b) Enlarged view of the red box in (a); (c) Enlarged view of the yellow box in (b); (d) Enlarged view of the blue box in (c). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

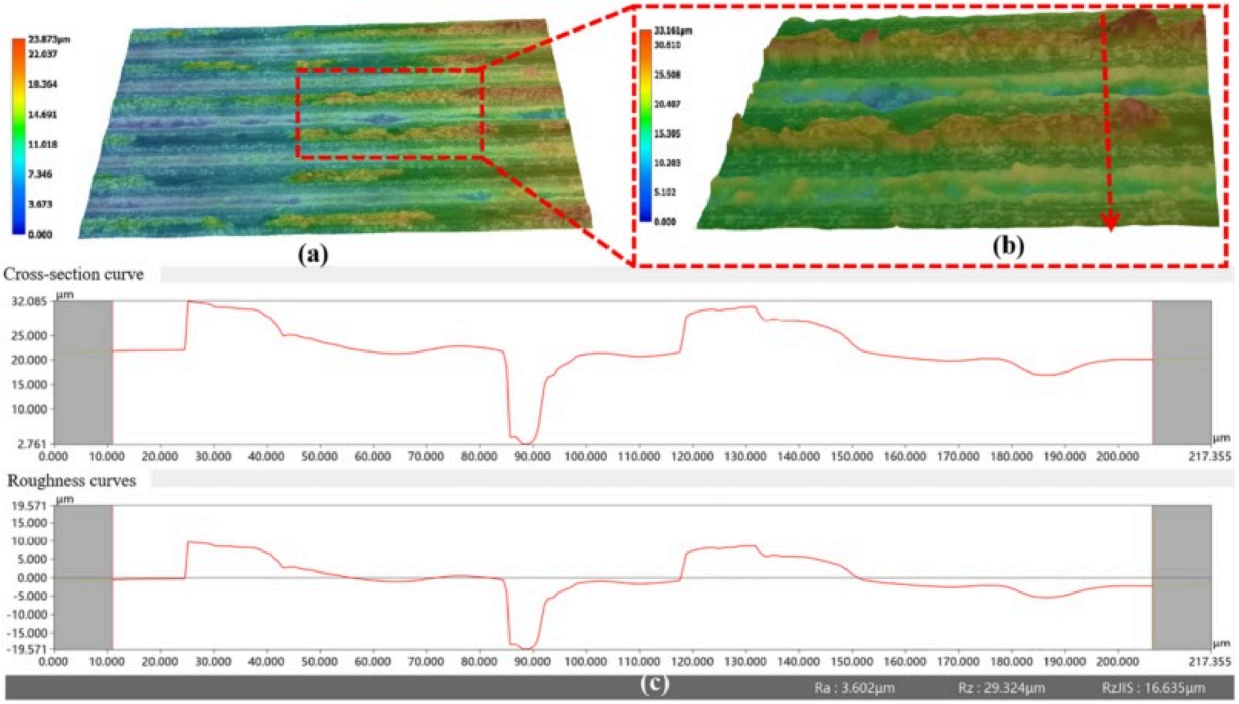
Fig. 13 presents the three-dimensional topography of shear fractures on the SiC ceramic side of sapphire-SiC ceramic joints produced by direct femtosecond laser microwelding under P=7.05 W and V=50 mm/s. As shown in Fig. 13(a), numerous strip-like sapphire regions are observed on the fracture surface of the SiC ceramic, indicating that the sapphire component near the interface has undergone fracture. Fig. 13(b–c) show that multiple hole structures containing individual sapphire nanoparticles exist between the strip-like sapphire regions on the fracture surface. The maximum depth of these holes is approximately 6 μm. This observation suggests that the fracture initiation occurred within the sapphire material itself, while blocky sapphire fragments with height ~30 μm remained adhered to the SiC ceramic due to the formation of reliable bonds. Fig. 14 illustrates the three-dimensional topography of shear fractures on the sapphire component in sapphire-SiC ceramic joints produced by direct femtosecond laser microwelding under P=7.05 W and V=50 mm/s. Notably, the remaining sapphire at the weld seam periphery demonstrates distinctive morphological characteristics. As shown in Fig. 14(b–c), multiple microvoids are present within the subsurface SiC ceramic matrix between adjacent SiC ceramic stripes on the fracture surface. The maximum depth of these voids is approximately 20 μm. This result indicates that fracture initiation occurred within the sapphire substrate, while prismoidal sapphire fragments with height ~33 μm remained securely retained on the SiC ceramic side through effective interfacial bonding.



Download: Download high-res image (510KB)

Download: Download full-size image

Fig. 13. 3D topography of SiC ceramics fracture surface of the joints welded at P=7.05W, V=50mm/s (top view). (a) 3D topography of SiC ceramics fracture surface; (b) Enlarged view of the red dashed box in (a); (c) Cross-section curve and roughness curve of the red dashed line in (b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

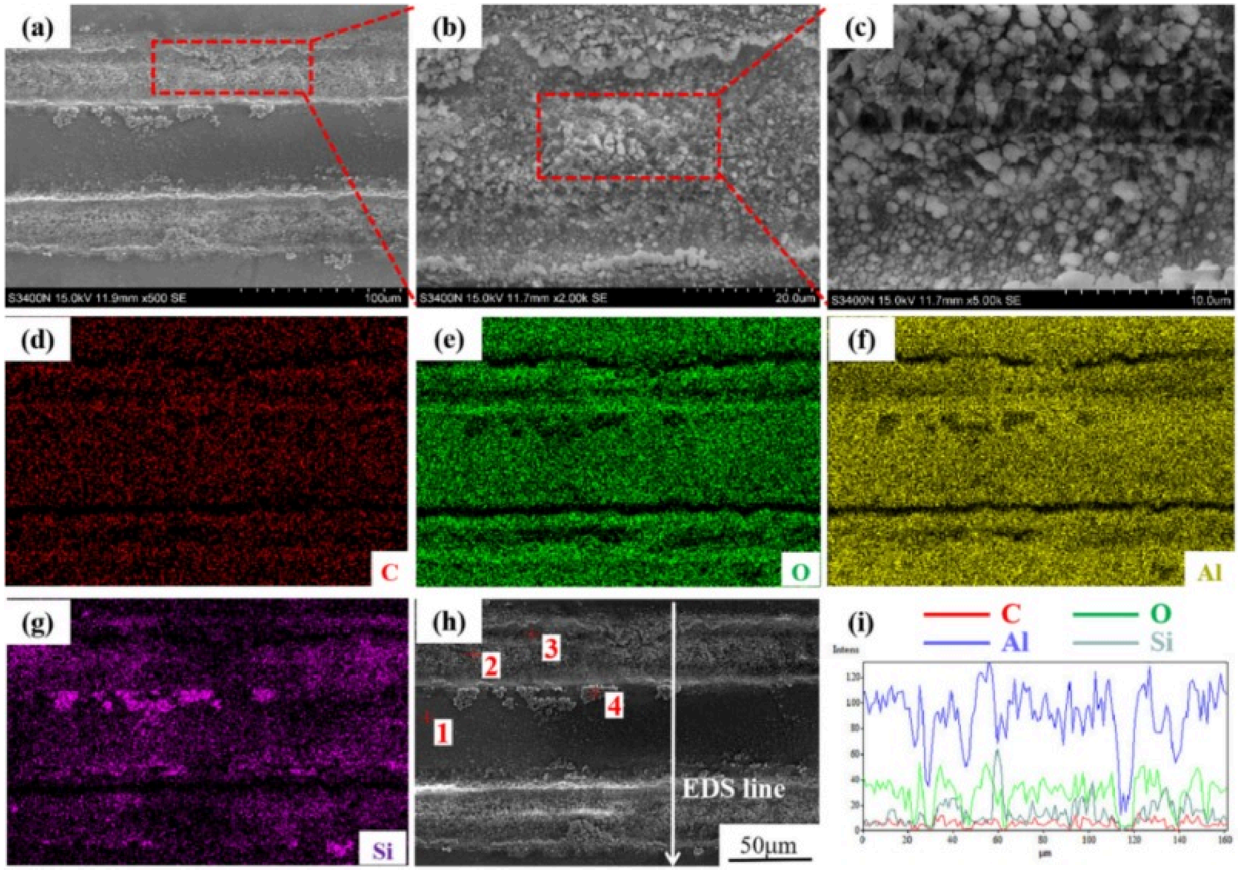


Download: Download high-res image (458KB)

Download: Download full-size image

Fig. 14. 3D topography of Sapphire fracture surface of the joints welded at P=7.05W, V=50mm/s (top view). (a) 3D topography of Sapphire fracture surface; (b) Enlarged view of the red dashed box in (a); (c) Cross-section curve and roughness curve of the red dashed line in (b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To elucidate the fracture mechanics, the shear fracture surfaces of sapphire-SiC ceramic joints were systematically characterized using SEM-EDS microanalysis (Fig. 15). Fig. 15(a) reveals periodically arranged sapphire stripes exhibiting identical welding trajectories on the sapphire component, suggesting that the sapphire side near the interface is fractured. Fig. 15(b–c) demonstrate extensive nanoparticle clusters distributed between adjacent sapphire stripes on the fracture surface. Fig. 15(d–j) confirms these particles possess heterogeneous composition comprising both SiC and Al_2O_3 nanoparticles. Notably, the droplet-like morphologies of these nanoparticles suggest sequential laser-induced plasma ablation followed by solid-state coalescence during rapid cooling. EDS quantification further shows aluminum enrichment in the SiC matrix adjacent to these nanoparticles, corroborating the laser-material interaction mechanism. The spherical features with smooth surface finish indicate complete melting of the material, while the absence of crystalline facets demonstrates amorphous phase formation during resolidification. SiC nanoparticles generated during laser ablation were found to facilitate micro-welding by enhancing local electromagnetic fields and promoting more efficient energy absorption at the interface. In addition, their presence in the molten pool contributes to localized thermal accumulation, which stabilizes the molten layer and aids in bridging micro-gaps. These combined effects improve the bonding quality and mechanical integrity of the joints.



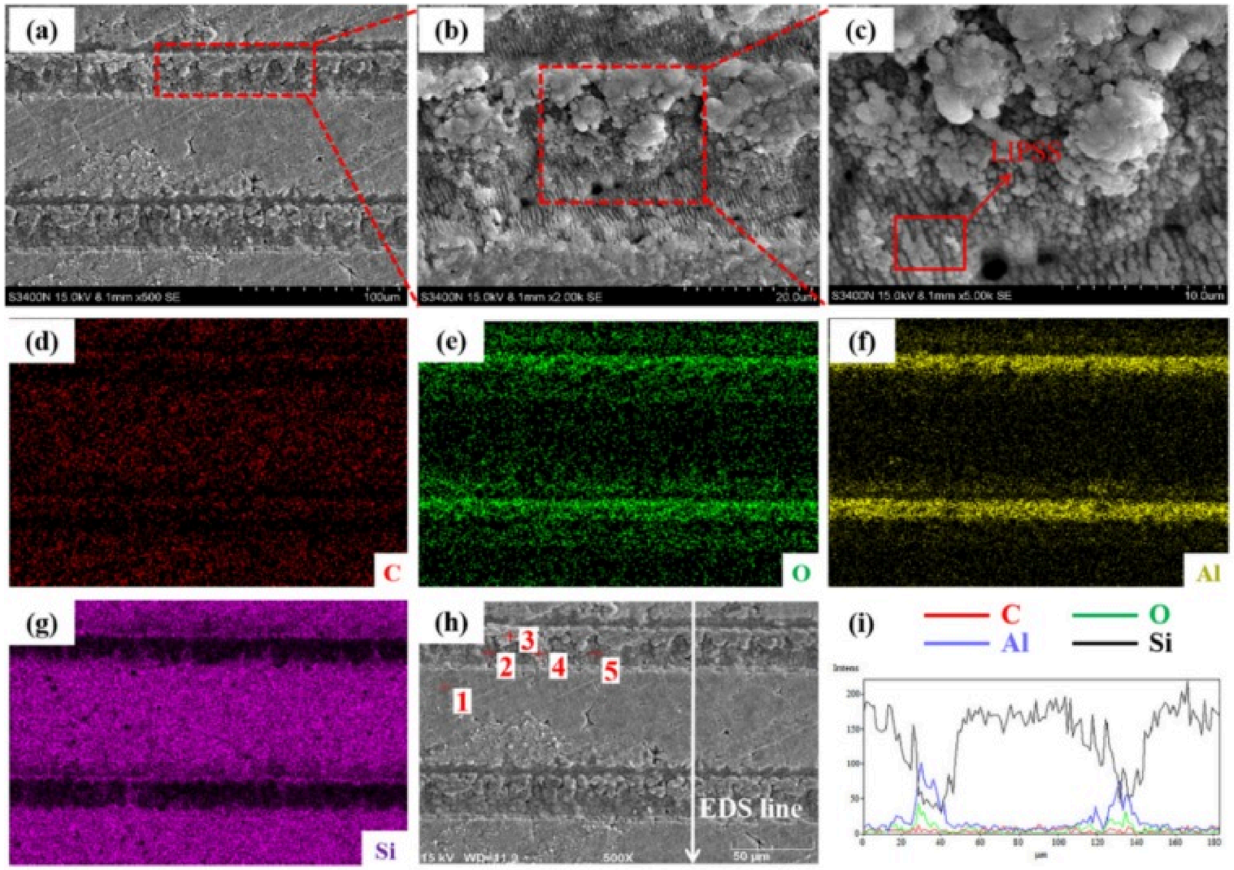
(j)

Test No.	Element(at.%)				Total
	C	O	Al	Si	
1	20.725	47.183	29.228	2.864	100
2	10.523	47.607	30.803	11.067	100
3	24.328	40.510	31.939	3.223	100
4	34.829	27.444	16.030	21.698	100

Download: Download high-res image (1MB)

Fig. 15. Morphologies of fractures of the sapphire surface after shear strength tests (top view) (a) SEM image; (b) Magnified areas marked with a red solid rectangle in (a); (c) Magnified areas marked with a red dotted rectangle in (b); (d-g) Element distribution in zone c in (a); (i) EDS line scanning results.; (j) EDS point scanning results. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 16 exhibits a predominant Al_2O_3-rich layer covering the welded seam, suggesting that the fracture initiation sites localized within the sapphire-SiC ceramic interface region. SEM observations of Fig. 16(b) reveal sapphire microparticles distributed bilaterally across the weld seam, likely originating from Al_2O_3 phase explosion during laser-material interaction. As shown in Fig. 16(c), cleavage facets and elongated dimples aligned with the shear loading direction dominate the primary fracture zone. The presence of cleavage planes indicates brittle fracture behavior, while the scarcity of refined dimples suggests limited plastic deformation prior to failure. This microstructural signature conclusively demonstrates the predominantly brittle nature of the fracture process. EDS mapping (Fig. 16(d–j)) reveals heterogeneous nano-particle compositions comprising both SiC and Al_2O_3. This spatial distribution pattern implies dual ablation mechanisms: upward ejection of SiC nanoparticles from the molten pool and downward precipitation of sapphire droplets through non-optical contact transfer under ultrafast cooling conditions. Notably, the lower right corner of Fig. 16(c) exhibits laser-induced periodic surface structures (LIPSS) formed exclusively on the SiC ceramic fracture surface. The appearance of LIPSS on SiC can be attributed to its strong absorption at 1030 nm and periodic modulation of the surface electromagnetic field under femtosecond laser irradiation, whereas sapphire mainly undergoes nonlinear absorption and microparticle ejection without forming ordered ripples. These nano-periodic ripples promote interfacial adhesion through mechanical interlocking and increased surface area, thereby enhancing the overall joint strength. In addition to plasma-mediated melting and nanoparticle-assisted absorption, the formation of laser-induced periodic surface structures (LIPSS) on the SiC ceramic surface further supports the bonding mechanism. As observed in Fig. 16(c), LIPSS appear as nano-periodic ripples distributed along the fracture surface of SiC. Their presence provides several benefits to micro-welding: (i) they significantly increase the effective interfacial surface area, which enhances adhesion strength; (ii) they introduce nano-scale mechanical interlocking that resists crack propagation along the joint; and (iii) the periodic modulation of the surface promotes localized electromagnetic field enhancement, thereby improving laser absorption and facilitating repeated melting during multi-scan irradiation. These effects collectively improve interfacial stability and strengthen the sapphire/SiC joint. Correlation with shear strength testing confirms that samples exhibiting pronounced LIPSS features reached shear strengths up to 30 MPa, suggesting a positive contribution of LIPSS to the overall bonding quality.



(j)

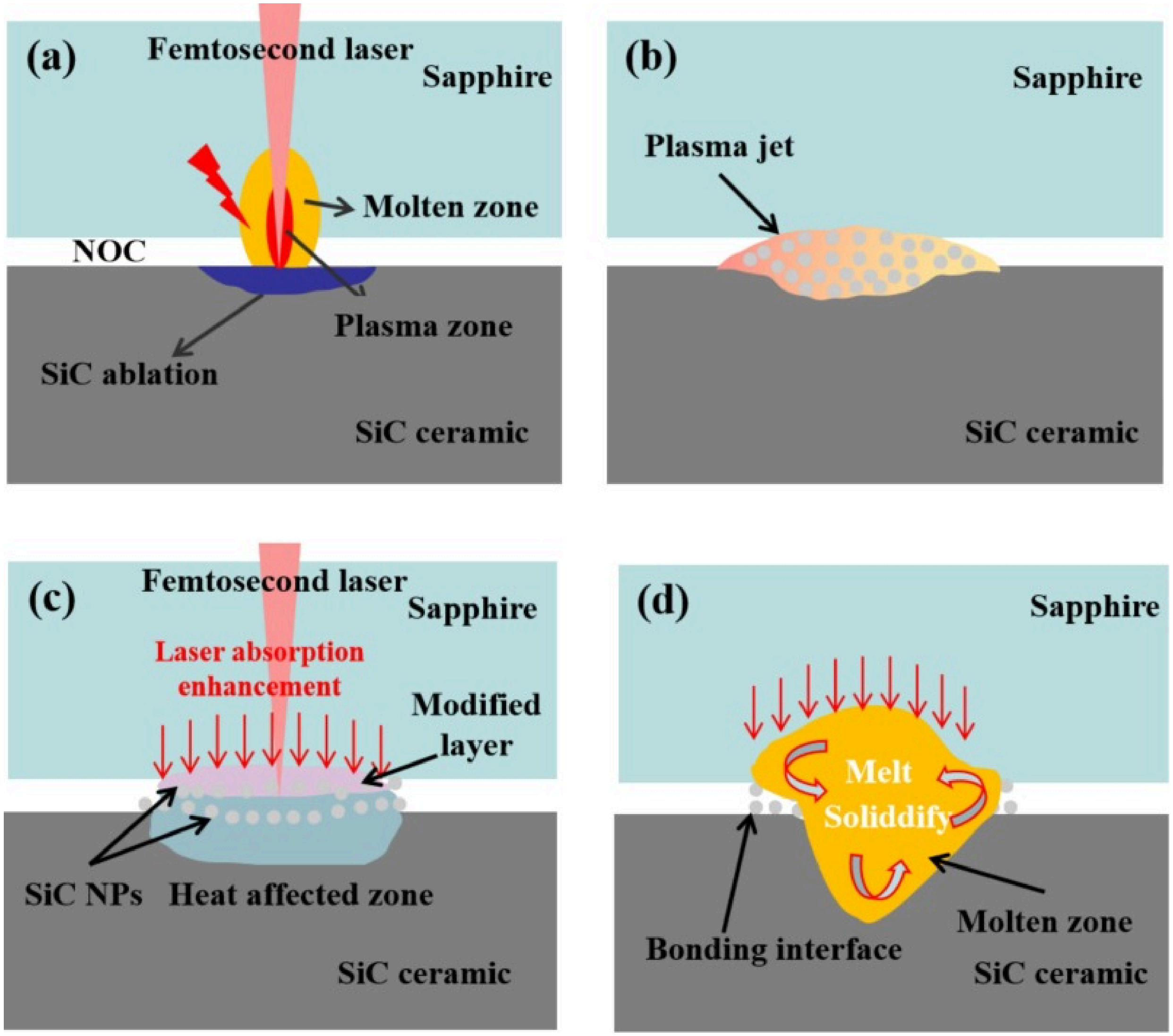
Test No.	Element(at.%)				Total
	C	O	Al	Si	
1	8.559	3.508	1.049	86.885	100
2	1.240	21.168	5.627	71.965	100
3	2.311	40.613	33.853	23.224	100
4	43.431	22.704	6.631	27.233	100
5	34.503	14.623	2.954	47.920	100

Fig. 16. Morphologies of fractures of the SiC ceramic surface after shear strength tests (top view) (a) SEM image; (b) Magnified areas marked with a red solid rectangle in (a); (c) Magnified areas marked with a red dotted rectangle in (b); (d–g) Element distribution in zone c in (a); (i) EDS line scanning results.; (j) EDS point scanning results. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In summary, EDSmapping revealed that Al, Si, O, and C redistribute across the sapphire/SiC interface after welding. This indicates that transient melting and ultrafast quenching enable non-equilibrium elemental diffusion and compound formation. Si released from SiC can react with oxygen species (originating from sapphire decomposition or ambient sources) to form SiO_2, which may subsequently combine with Al_2O_3 from sapphire to yield amorphous or crystalline Al-Si-O phases. Simultaneously, excess carbon may either precipitate as nanoparticles or form oxycarbides under localized oxidation. This chemically mixed transition layer not only bridges the lattice mismatch between sapphire and SiC but also enhances mechanical integrity through increased adhesion and interfacial toughness.

3.6. Formation process and mechanism of sapphire/SiC ceramics joint

The ultra-high energy density of femtosecond laser pulses induces resonant absorption in the material, causing vaporization with multi-stage energy coupling: absorbed thermal energy promotes plasma ionization through collisional processes, thereby generating a high-temperature plasma cloud between the laser beam and workpiece surface. Because of its high temperature and density, the plasma absorbs and scatters the laser energy, hindering the beam's transmission to the material. Microstructural investigations of sapphire-SiC ceramic joints combined with energy balance modeling reveals: plasma formation initiates when laser power density exceeds $10^{14} W/cm^2$ threshold through nonlinear absorption processes. The plasma-induced shielding effect creates self-regulating feedback loop: increased plasma opacity reduces laser energy deposition, which in turn lowers the plasma generation rate until steady-state equilibrium is achieved. The femtosecond laser-material interaction mechanism is dominated by nonlinear absorption-induced melting and rapid solidification dynamics. As depicted in Fig. 17(a), the sapphire/SiC interface achieves a non-optical-contact(NOC) configuration during transmission welding. Here, the femtosecond laser beam propagates through the sapphire substrate and directly couples with the SiC ceramic surface. Photon energy absorption initiates through the inverse bremsstrahlung mechanism [33], where electrons first capture laser energy before transferring it to the lattice via electron-phonon scattering. This intense energy deposition triggers plasma formation comprising ionized species and neutrals, which are ejected through Coulomb explosion and thermal vaporization [34]. The expanding plasma interacts with the sapphire confinement layer (Fig. 17(a)), inducing localized phase transformations and selective material ablation. Notably, the modified interfacial layer and deposited nanoparticles create secondary energy absorption centers. These sites enable surface plasmon resonance between sapphire and SiC nanoparticles [35], generating localized electromagnetic field enhancements that amplify energy coupling efficiency. Furthermore, sustained heat accumulation from high-repetition-rate laser pulsing promotes nanoparticle coalescence and improves metal-sapphire wettability, ultimately stabilizing the sapphire/SiC joint formation (Fig. 17(d)). SEM characterization of the optimized weld interface reveals homogeneous fusion between the ceramic components. Mechanical testing demonstrates superior joint integrity, with fracture surfaces propagating through both parent materials and the weld center, as evidenced by shear strength measurements and fracture morphology analysis.



Download: Download high-res image (375KB)

Download: Download full-size image

Fig. 17. Schematic diagrams of the joint formation during femtosecond laser microwelding of sapphire and SiC ceramics (a) SiC ablation; (b) Plasma excitation and expansion; (c) Absorption enhancement based on the micro-defects and nanoparticles deposition; (d) Melting and solidification for joint formation.

A multi-scan femtosecond laser method effectively achieves welding strength up to 30 MPa for non-optical contact samples, differing from conventional optical contact welding. The process involves three main steps: (1) In the first scan, the laser is focused below the lower sample surface, forming a modified region through nonlinear absorption. Multiple scans elevate this region, enabling bonding between samples. (2) In the second scan, the solidified region is ablated, with some material deposited onto the upper sample, gradually forming a power deposition layer and relieving internal stresses. (3) During subsequent (Nth) scans, the deposition layer thickens and connects with the modified region, eventually creating a new bonding zone. Ejected particles are re-melted, and repeated scanning acts as post-treatment, reducing thermal stress and improving joint quality [36]. In summary, the bonding mechanism of sapphire/SiC ceramics under femtosecond laser irradiation can be understood as a multi-stage process. Initial pulses create a modified interface layer via nonlinear absorption, followed by plasma-assisted melting and nanoparticle deposition. With increasing pulse number, repeated scanning promotes interfacial mixing, reduces thermal stresses, and progressively enlarges the bonding zone. Ultimately, the coalescence of modified and deposited layers forms a stable joint with enhanced strength.

4. Conclusion

This work demonstrates that femtosecond laser multi-scan micro-welding enables reliable direct bonding of sapphire and rough SiC ceramics under natural stacking without optical contact or external pressure. Ultrafast imaging and systematic experiments reveal that plasma dynamics, nanoparticle-assisted absorption, and the formation of LIPSS synergistically improve interfacial bonding, achieving shear strengths up to 30 MPa. The key findings are summarized below:

1) SiC ceramic surfaces were left unpolished to establish a non-optical contact (NOC) interface with sapphire. Under femtosecond laser irradiation, rapid ionization within the interface produced a high-density plasma that

triggered ultrafast localized melting, interdiffusion, and the radial dispersion of Al_2O_3 particulates.

2) Utilizing an ultrafast pump-probe shadowgraphy system, the phenomenon of plasma splashing during the sapphire-ceramic micro-welding process was captured. The results reveal that the self-focusing effect and nonlinear absorption of sapphire play a pivotal role in plasma generation, significantly influencing the shear strength of the joints.

3) The influence of processing parameters on joint characteristics was systematically investigated, with particular focus on laser power density and scanning velocity. Experimental results revealed a non-monotonic relationship between joint strength and processing parameters: as either laser power or scanning speed increased, the joint strength exhibited a characteristic parabolic trend, reaching an optimum value before subsequent decline.

4) A laser-induced periodic surface structure (LIPSS) was formed at the sapphire-side fracture of the femtosecond laser microwelded sapphire and silicon carbide ceramics. This periodic surface structure facilitates the bonding of sapphire and silicon carbide ceramics.

This highly efficient and straightforward novel FSLMW welding technology for sapphire-ceramic holds great potential for future practical applications. Future work will incorporate numerical simulations to quantitatively validate the role of nanoparticles in femtosecond laser welding. In future work, we seek to proceed the thermal fatigue test, dynamic and static fatigue test, and other mechanical property tests of the sapphire-SiC ceramic joint, and the sapphire-SiC ceramic products will be investigated in real application scenarios.

CRediT authorship contribution statement

Changhao Ji: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ziyue Yu:** Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Jindou Wu:** Writing – review & editing, Visualization, Investigation, Data curation. **Qijian Zhu:** Writing – review & editing, Investigation, Funding acquisition, Formal analysis. **Chengyun Wang:** Writing – review & editing, Validation, Methodology, Formal analysis. **Liming Lei:** Writing – review & editing, Visualization, Methodology. **Hongchao Qiao:** Writing – review & editing, Validation, Data curation. **Yu Long:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key Research and Development Program of China (Grant No. 2022YFB4601601), the Key R&D Program of Guangxi Province (Grant No. GKAB23026101), Internet Information Innovation Talent of Guangxi Province (Grant No. 202500081), and the Guangxi Natural Science Foundation (Grant No. 2023GXNSFBA026287).

Appendix A. Supplementary data

Download: Download Word document (16MB)

Supplementary figures

References

[1] S. Zhang, J. Lin, C. Wang, *et al.*
Improvement of microstructure and mechanical properties of sapphire/Kovar alloy joints brazed with Ag-Cu-Ti composite filler by adding B powders
J Manuf Process, 99 (2023), pp. 243-253, 10.1016/j.jmapro.2023.05.050 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[2] L. He, C. Chen, H. Li, *et al.*
Research advances in joining processes of sapphire
Int J Adv Manuf Technol, 121 (1) (2022), pp. 59-81, 10.1007/s00170-022-09199-9 ↗
View in Scopus ↗ Google Scholar ↗

[3] M. Trunec, J. Klimke, Z.J. Shen
Transparent alumina ceramics densified by a combinational approach of spark plasma sintering and hot isostatic pressing
J Eur Ceram Soc, 36 (16) (2016), pp. 4333-4337, 10.1016/j.jeurceramsoc.2016.06.004 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[4] Chandra P. Khattak, Raj Shetty, C. Richard Schwerdtfeger, *et al.*
World's largest sapphire for many applications
J Cryst Growth, 452 (2016), pp. 44-48, 10.1016/j.jcrysgro.2015.11.026 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[5] C.T. Sun, D.F. Xue
Single crystal growth mechanism of sapphire
Mater Technol, 28 (5) (2013), pp. 286-289, 10.1179/1753555713Y.0000000060 ↗
View in Scopus ↗ Google Scholar ↗

[6] J. Liu, X. Shi, L. Jiang, *et al.*
Progress in transparent bulletproof armorbased on ceramics

J Mater Eng, 49 (11) (2021), pp. 30-40
https://link-cnki-net.kmclibproxy.khu.ac.kr/urlid/11.1800.TB.20211104.1052.032 ↗

Crossref ↗ View in Scopus ↗ Google Scholar ↗

[7] C. Li, Z. Xie, L. Zhao
Research and application of sintering technologies for SiC ceramic materials: a review
J Ceram, 41 (02) (2020), pp. 137-149, 10.13957/j.cnki.tcxb.2020.02.001 ↗

View PDF View article Google Scholar ↗

[8] Y. Li, F. Huang, Z. Liang
Properties and applications of silicon carbide ceramics
Ceramics, 05 (2007), pp. 36-41, 10.19397/j.cnki.ceramics.2007.05.009 ↗

Google Scholar ↗

[9] Lei Danggang
Analysis of laser welding cracks on kovar alloy box
Electron Process Technol, 33 (1) (2012), pp. 45-49, 10.14176/j.issn.1001-3474.2012.01.001 ↗

Google Scholar ↗

[10] F. Niklaus, G. Stemme, J.Q. Lu, *et al.*
Adhesive wafer bonding
J Appl Phys, 99 (3) (2010), 10.1063/1.2168512 ↗
031101-031101-28

Google Scholar ↗

[11] L. Lin
MEMS post-packaging by localized heating and bonding
IEEE Trans Adv Packag, 23 (4) (2000), pp. 608-616, 10.1109/6040.883749 ↗

View in Scopus ↗ Google Scholar ↗

[12] Y.T. Cheng, L. Lin, K. Najafi

Localized silicon fusion and eutectic bonding for MEMS fabrication and packaging

J Microelectromech Syst, 9 (1) (2000), pp. 3-8, 10.1109/84.825770 ↗

View in Scopus ↗ Google Scholar ↗

[13] K. Cvecek, S. Dehmel, I. Miyamoto, *et al.*

A review on glass welding by ultra-short laser pulses

Int J Extreme Manuf, 1 (4) (2019), Article 042001, 10.1088/2631-7990/ab55f6 ↗

View in Scopus ↗ Google Scholar ↗

[14] G. Zhang, G. Cheng

Direct welding of glass and metal by 1 kHz femtosecond laser pulses

Appl Opt, 54 (30) (2015), pp. 8957-8961, 10.1364/AO.54.008957 ↗

View in Scopus ↗ Google Scholar ↗

[15] R.M. Carter, M. Troughton, J. Chen, *et al.*

Towards industrial ultrafast laser microwelding: SiO2 and BK7 to aluminum alloy

Appl Opt, 56 (16) (2017), pp. 4873-4881, 10.1364/AO.56.004873 ↗

View in Scopus ↗ Google Scholar ↗

[16] R.M. Carter, M. Troughton, J. Chen, *et al.*

Picosecond laser welding of optical to metal components [C]//laser-based Micro-and Nanoprocessing X

SPIE, 9736 (2016), pp. 244-253, 10.1117/12.2213227 ↗

Google Scholar ↗

[17] O.P. Ciuca, R.M. Carter, P.B. Prangnell, *et al.*

Characterisation of weld zone reactions in dissimilar glass-to-aluminium pulsed picosecond laser welds

Mater Charact, 120 (2016), pp. 53-62, 10.1016/j.matchar.2016.08.013 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[18] C. Ji, Y. Huang, X. Chen, *et al.*
Direct microwelding of dissimilar glass and Kovar alloy without optical contact using femtosecond laser pulses
J Cent South Univ, 29 (10) (2022), pp. 3422-3435, 10.1007/s11771-022-5091-9 ↗
View in Scopus ↗ Google Scholar ↗

[19] Rui Pan, Dong Yang, Taoshuai Zhou, Yinghao Feng, Zhisen Dong, Zhaoyang Yan, *et al.*
Micro-welding of sapphire and metal by femtosecond laser
Ceram Int, 49 (13) (2023), pp. 21384-21392, 10.1016/j.ceramint.2023.03.267 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[20] M. Chambonneau, Q. Li, V.Y. Fedorov, *et al.*
Taming ultrafast laser filaments for optimized semiconductor–metal welding
Laser Photonics Rev, 15 (2) (2021), Article 2000433, 10.1002/lpor.202000433 ↗
View in Scopus ↗ Google Scholar ↗

[21] G. Zhang, J. Bai, W. Zhao, *et al.*
Interface modification based ultrashort laser microwelding between SiC and fused silica
Opt Express, 25 (3) (2017), pp. 1702-1709, 10.1364/OE.25.001702 ↗
View in Scopus ↗ Google Scholar ↗

[22] I. Miyamoto, Y. Okamoto, A. Hansen, *et al.*
High speed, high strength microwelding of Si/glass using ps-laser pulses
Opt Express, 23 (3) (2015), pp. 3427-3439, 10.1364/OE.23.003427 ↗
View in Scopus ↗ Google Scholar ↗

[23] I.H.W. Nordin, Y. Okamoto, A. Okada, *et al.*
Effect of wavelength and pulse duration on laser micro-welding of monocrystalline silicon and glass
Appl Phys A Mater Sci Process, 122 (2016), pp. 1-7, 10.1007/s00339-016-9849-5 ↗

Google Scholar ↗

[24] J. Zhang, S. Xu, Y. Dong, *et al.*
Microwelding of glass to silicon by green ultrafast laser pulses
Opt Laser Technol, 120 (2019), Article 105720, 10.1016/j.optlastec.2019.105720 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[25] E.H. Penilla, L.F. Devia-Cruz, A.T. Wieg, *et al.*
Ultrafast laser welding of ceramics
Science, 365 (6455) (2019), pp. 803-808, 10.1126/science.aaw6699 ↗
View in Scopus ↗ Google Scholar ↗

[26] K. Itoh, Y. Ozeki
Novel processes in laser microfabrication and microjoining
Proceedings of the 29th International Congress on Applications of Lasers and Electro-Optics, ICALEO 2010 -Congress Proceedings, F, 1064 (2010), pp. 1064-1068, 10.2351/1.5061940 ↗
(2010)
View in Scopus ↗ Google Scholar ↗

[27] H. Yuan, C. Li, B. Yang, *et al.*
Direct welding of diffuse alumina ceramics by ultrashort pulse laser
J Eur Ceram Soc, 44 (1) (2024), pp. 574-578, 10.1016/j.jeurceramsoc.2023.08.034 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[28] H. Jiang, C. Li, X. Mao, *et al.*
Reliable joining between $MgAl_2O_4$ and Ti6Al4V by ultrashort pulse laser
Ceram Int, 50 (2024), pp. 31609-31617, 10.1016/j.ceramint.2024.05.468 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[29] T. Tamaki, W. Watanabe, J. Nishii, K. Itoh
Welding of transparent materials using femtosecond laser pulses
Jpn J Appl Phys, 44 (22) (2005), pp. 687-689, 10.1143/JJAP.44.L687 ↗

Google Scholar ↗

[30] K. Cvecek, R. Odato, S. Dehmel, I. Miyamoto, M. Schmidt
Gap bridging in joining of glass using ultra short laser pulses
Opt Express, 23 (5) (2015), pp. 5681-5693, 10.1364/OE.23.005681 ↗
View in Scopus ↗ Google Scholar ↗

[31] A. Braun, G. Korn, X. Liu, D. Du, J. Squier, G. Mourou
Self-channeling of high-peak-power femtosecond laser pulses in air
Opt Lett, 20 (1) (1995), Article 73, 10.1364/OL.20.000073 ↗
View in Scopus ↗ Google Scholar ↗

[32] T. Yan, L. Ji
Ultrafast laser filamentation in transparent solids
Ultrafast Sci, 3 (2023), Article 0023, 10.34133/ultrafastscience.0023 ↗
View in Scopus ↗ Google Scholar ↗

[33] K.C. Phillips, H.H. Gandhi, E. Mazur, S.K. Sundaram
Ultrafast laser processing of materials: a review
Adv Opt Photon, 7 (2015), pp. 684-712, 10.1364/AOP.7.000684 ↗
View in Scopus ↗ Google Scholar ↗

[34] P. Sankar, H.D. Shashikala, R. Philip
Effect of laser beam size on the dynamics of ultrashort laser-produced aluminum plasma in vacuum
Phys Plasmas, 26 (2019), Article 013302, 10.1063/1.5054195 ↗
View in Scopus ↗ Google Scholar ↗

[35] J. Zhang, D. Zhu, J. Yan, C.A. Wang
Strong metal-support interactions induced by an ultrafast laser
Nat Commun, 12 (2021), p. 6665, 10.1038/s41467-021-27000-5 ↗
View in Scopus ↗ Google Scholar ↗

[36] X.S. Jia, K. Li, Z. Li, *et al.*
Multi-scan picosecond laser welding of non-optical contact soda lime glass
Opt Laser Technol, 161 (2023), Article 109164, 10.1016/j.optlastec.2023.109164 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

View Abstract

Recommended articles

Based on reading popularity

Research article Full text access

Direct joining of YAG crystal and Cu by femtosecond laser

Haoran Yang, ..., Jian Cao

Journal of Manufacturing Processes • Volume 152 • 2025 • Pages 114-125

View PDF

Research article Full text access

Microwelding of glass to silicon by green ultrafast laser pulses

Jiejuan Zhang, ..., Qingmao Zhang

Optics & Laser Technology • Volume 120 • 2019 • Article 105720

View PDF

Research article Full text access

Maskless fabrication of micro-gradient patterned ion density enhanced moisture-driven power generation using a phase spatially shaped femtosecond laser

Xiaozhe Chen, ..., Lan Jiang

Journal of Manufacturing Processes • Volume 135 • 2025 • Pages 1-9

View PDF

Research article Full text access

Direct welding of silicon carbide and fused silica using femtosecond lasers: Effects of repetition rate and focal depth

Yu-Guo Jiang, ..., Yasuhiro Okamoto

Optics & Laser Technology • Volume 184 • 2025 • Article 112489

View PDF

Research article Full text access

Direct joining of quartz glass and copper by nanosecond laser

Yinghao Feng, ..., Shujun Chen

Ceramics International • Volume 49, Issue 22, Part B • 2023 • Pages 36056-36070

View PDF

Research article Open access

The characteristics and dynamics of fused silica-aluminum alloy welding during mJ-level femtosecond laser

Lin Zhang, ..., Xiuquan Ma

Materials & Design • Volume 239 • 2024 • Article 112790

View PDF

Cited by (1)

Research article

Novel double-sided laser welding approach for strength enhancement and defect mitigation in wide-gap glass joints

Jiang N., ..., Wang C.

Optics and Laser Technology • Volume 201 • 2026 • Article 115200

Show abstract

Metrics

Citations